Convolutional Neural Networks Part 1

Agenda

- Motivation
- Convolutional Layer
- PyTorch Implementation
- More on CNNs

How were the last four weeks?

Q: How is the pace of the lectures? Difficulty level?

OK

Q: What do you think of the assignment? How's the difficulty level?

LONG

Q: How are the Q/A Sessions going?

We welcome your feedback. Please reach out if you have suggestions.

Recap: Motivating Problem -1

- Car vs Semi-Truck Classifier

Input: image

?

Output: pick the correct categories

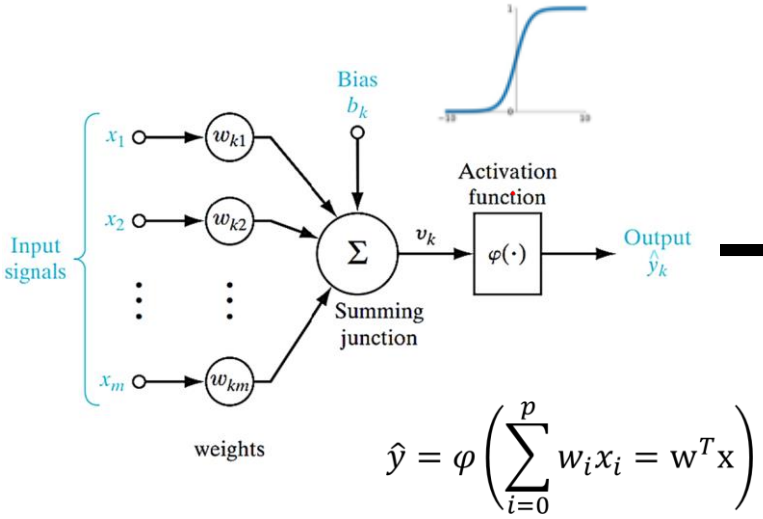
1 - car

0 - truck

Recap: Motivating Problem -2

1. Dataset: CIFAR
2. Model: Logistic Regression (simple linear network)

Input: image [1 x 1024]



$$\hat{y} = \varphi\left(\sum_{i=0}^{p} w_i x_i = \mathrm{w}^T \mathrm{x}\right)$$

Output: pick the correct categories

1 - car

0 - truck

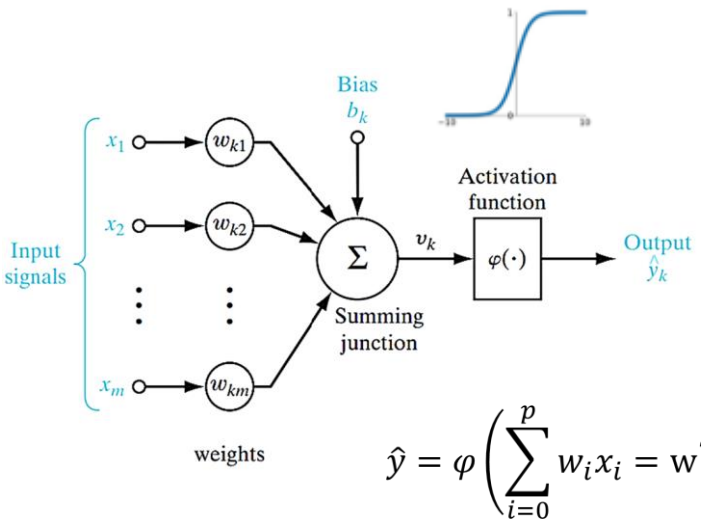
Accuracy: 65%

Recap: Motivating Problem -3

- Logistic Regression + Feature Engineering

Input:

[length, width, height, size] 1 x 4



Output: pick the correct categories

1 - car

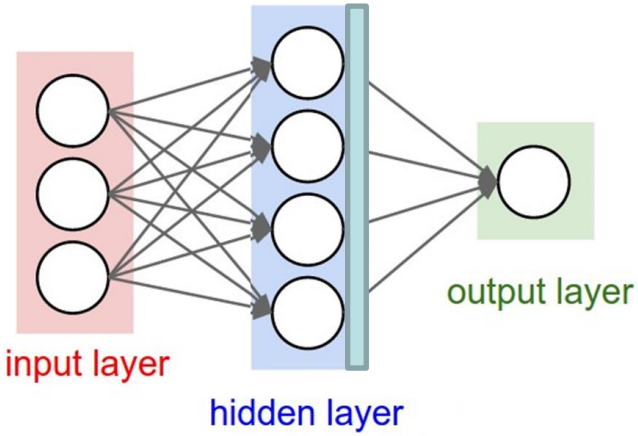
0 - truck

Accuracy: 99%

Recap: Motivating Problem -4

- Multi-Layer Perceptron (MLP) to automate feature selection (i.e., nonlinear neural network)

Input: image [1 x 1024]



Output: pick the correct categories

1 - car

0 - truck

Accuracy: ~74%

Recap: Motivating Problem -4

- Q: Can we do better? And how?

Assume we are using only MLP

Input: image

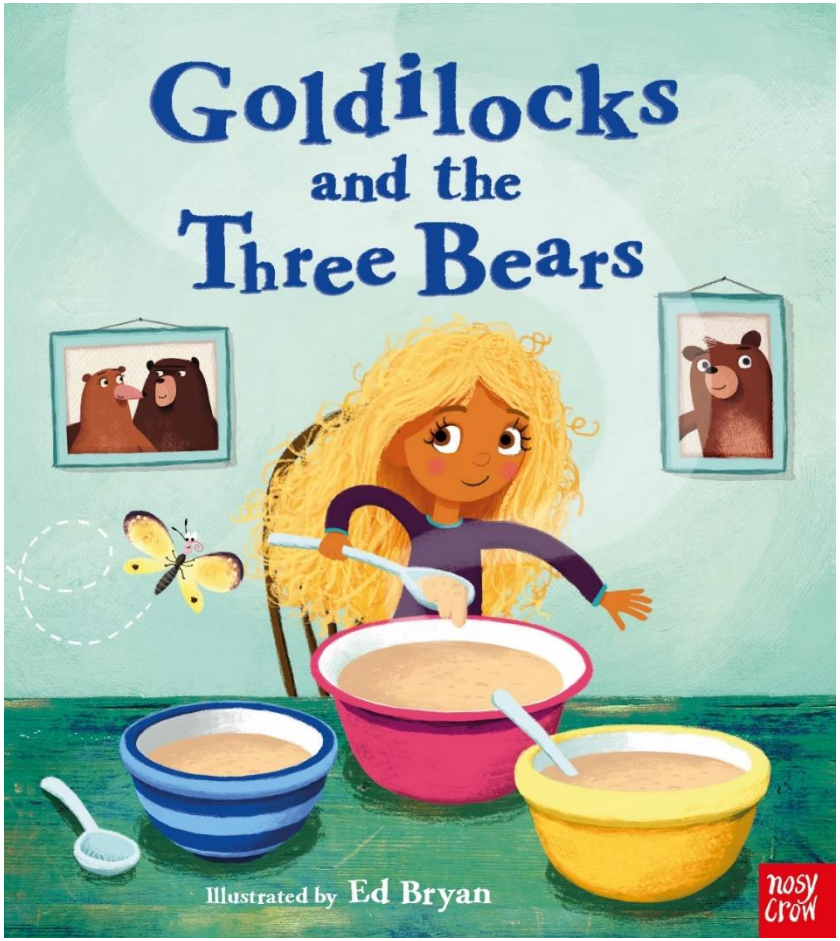
?

Output: pick the correct categories

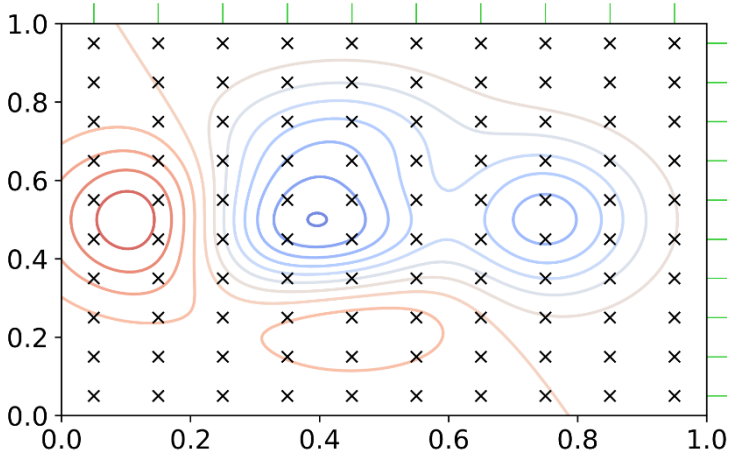
1 - car

0 - truck

1. Hyperparameter Tuning

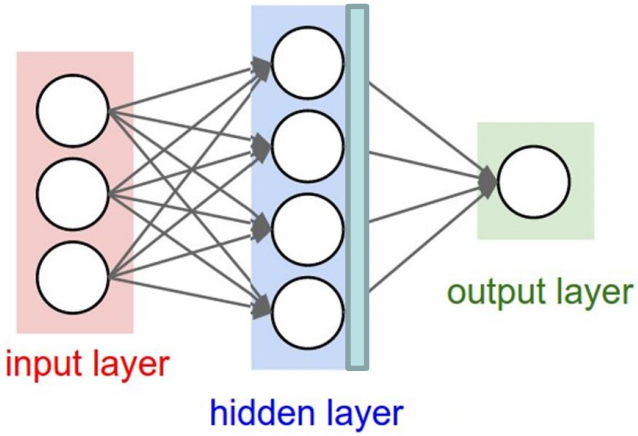


- Neural networks require a great deal of time tuning hyperparameters
- Practice makes perfect



2. Use Higher Resolution Images

Input: image [200 x 200]



Output: pick the correct categories

1 - car

0 - truck

- This will work, but there are some challenges to overcome…

PyTorch Example: # of Parameter

batch_size, # channels, height, width

images in pytorch → NCHW

```
class Classifier(nn.Module):
    def __init__(self):
        super(Classifier, self).__init__()
        self.layer1 = nn.Linear(32 * 32, 50)
        self.layer2 = nn.Linear(50, 20)
        self.layer3 = nn.Linear(20, 1)
    def forward(self, img):
        x = img.view(-1, 32 * 32)
        x = F.relu(self.layer1(x))
        x = F.relu(self.layer2(x))
        x = self.layer3(x)
        return x
```

find shape N,1,32,32

N,1,1024 N,1,50 N,1,20 N,1,1

Q: How many layers does this network have? 3

Q: How many weights are in the first layer of this network?

bias

param1 = (32·32)·50 + 50

+ = (32·32+1)·50

Q: How many total parameters are there?

param2 = (50+1)·20

+

param3 = (20+1)·1

=

High Resolution Image

200 pixels

200 pixels

What if our network is bigger?

Example:

- Input image: 200 × 200 pixels
- First hidden layer: 500 units
- Second hidden layer: 200 units

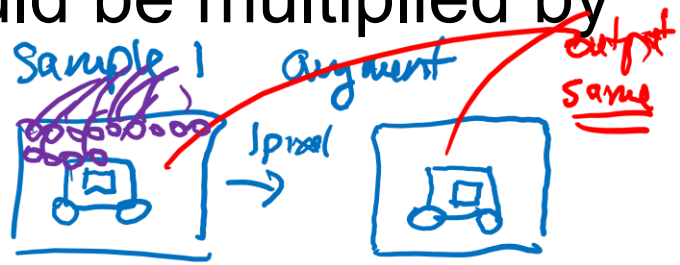
Q: How many parameters are there?

~20 million

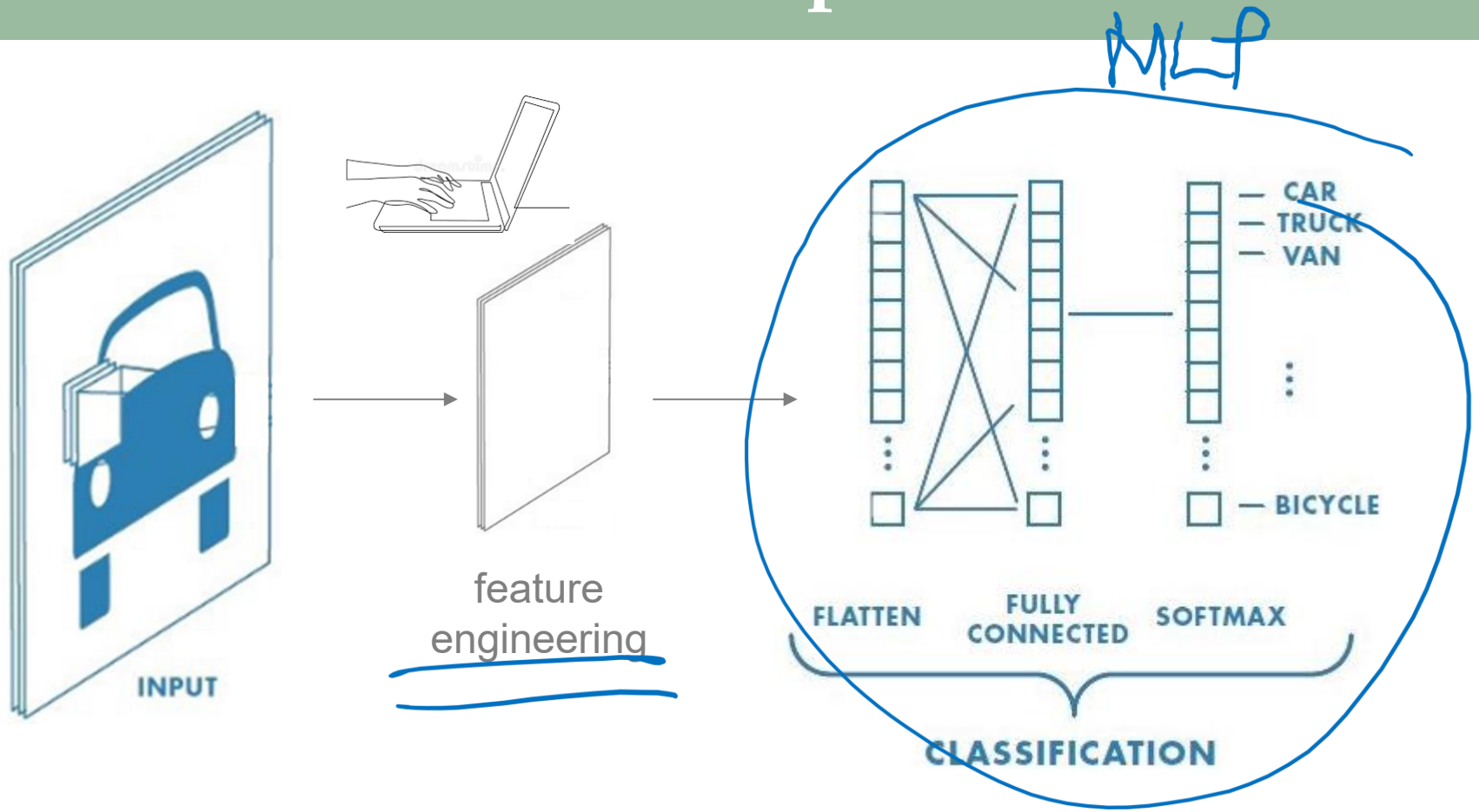
Q: Why would using large fully connected layers be problematic?

What if our network is bigger?

- Q: Why would using large fully connected layers be problematic?
  - slow training due to large number of parameters
  - requires a lot of training data to avoid overfitting (exponential relationship with # of parameters)
  - does not make use of spatial relationships
  - small shift in image can result in large change in prediction (i.e., each pixel would be multiplied by a different parameter/weight)



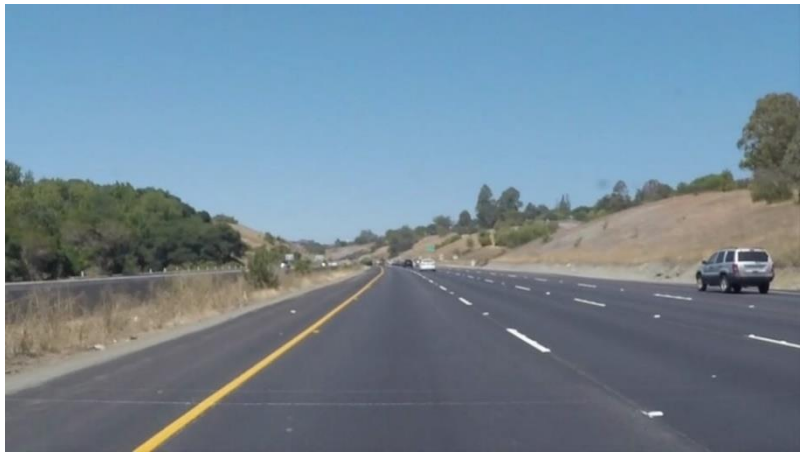
What was done in the past?

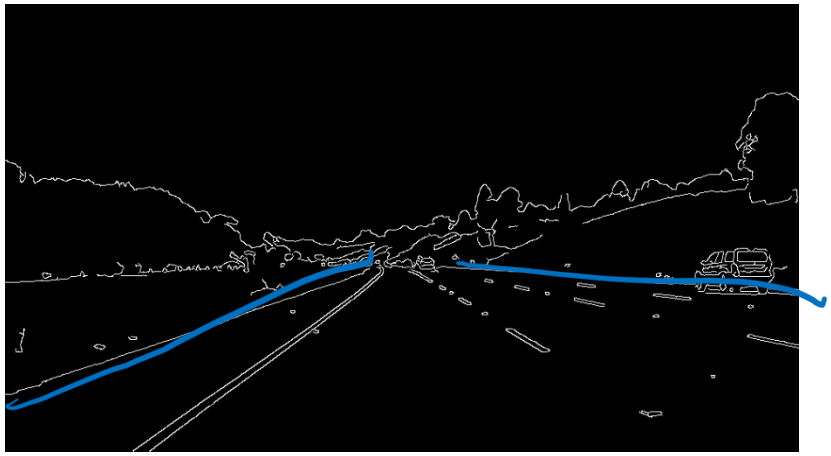


Most of the effort went into feature engineering or processing of the input.

Convolutional Filters

- Engineers have used the idea of **convolutional filters in computer vision** even before the rise of machine learning.
- They hand-coded filters that can detect simple features, for example to **detect edges** in various orientations.





Convolution in 2-D for Images

Convolution of Image **I** with filter kernel **K**:

- Flip rows and columns of kernel
- Multiply each pixel in range of kernel by the corresponding element of flipped kernel, sum all these products and write to a new 2d array.
- Slide kernel across all areas of the image until you reach the ends.
  - Optional, can use zero padding to maintain size of output

K

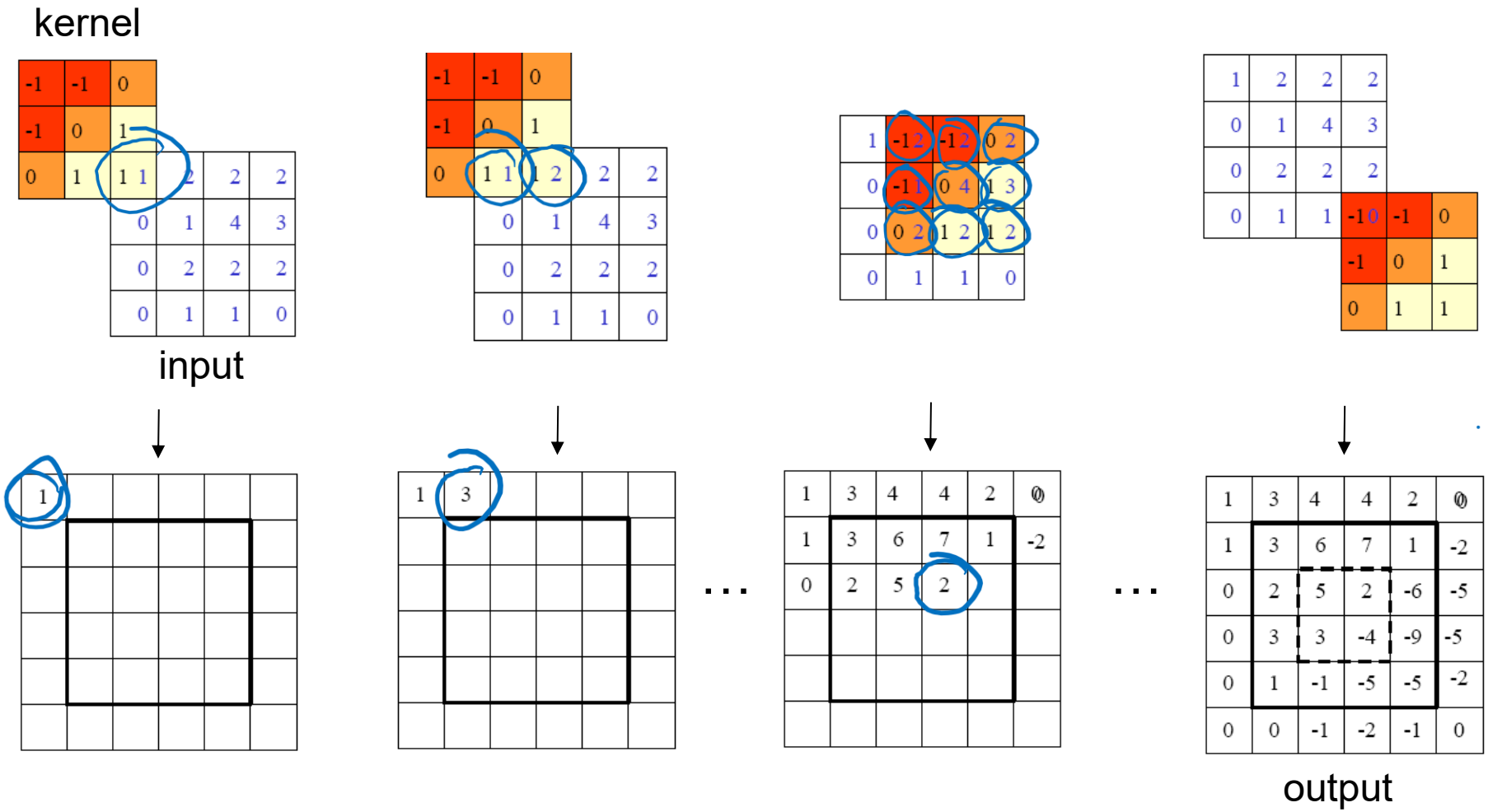
Kernel

1	1	0
1	0	-1
0	-1	-1

Image **I**

1	2	2	2
0	1	4	3
0	2	2	2
0	1	1	0

Demonstration of 2D Convolution



Blurring Filter - Averaging

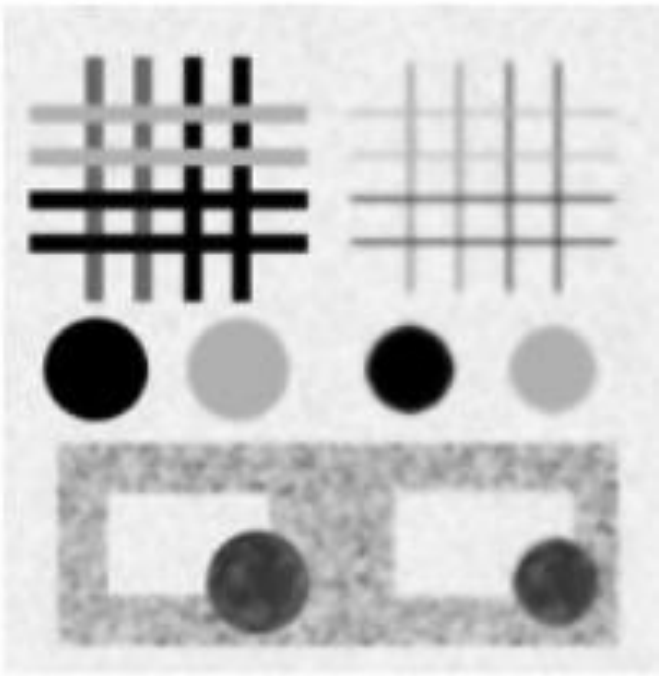
- Blurring averages out pixel intensities in an image.
- It is a form of low pass filter
- Attenuates high frequency features resulting from sharp transitions in the image

kernel

$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Sobel Filter X

Q: What does this filter do?



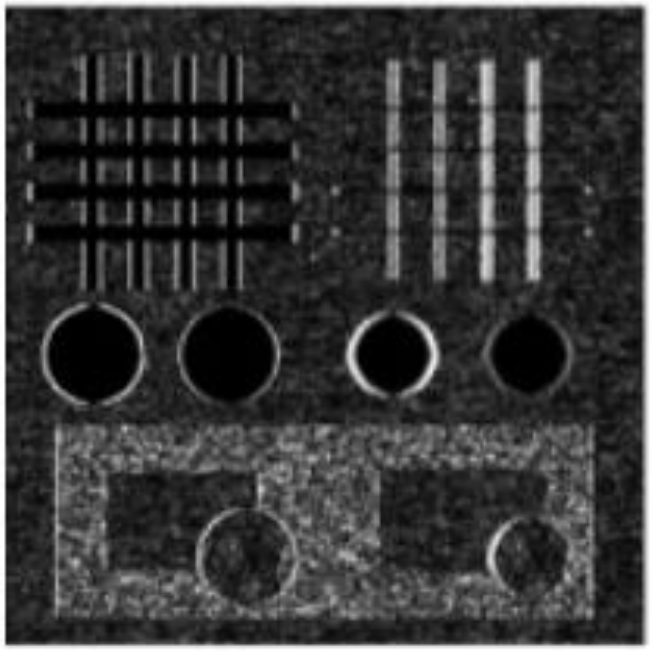
Original

*

kernel

-1	0	1
-2	0	2
-1	0	1

=

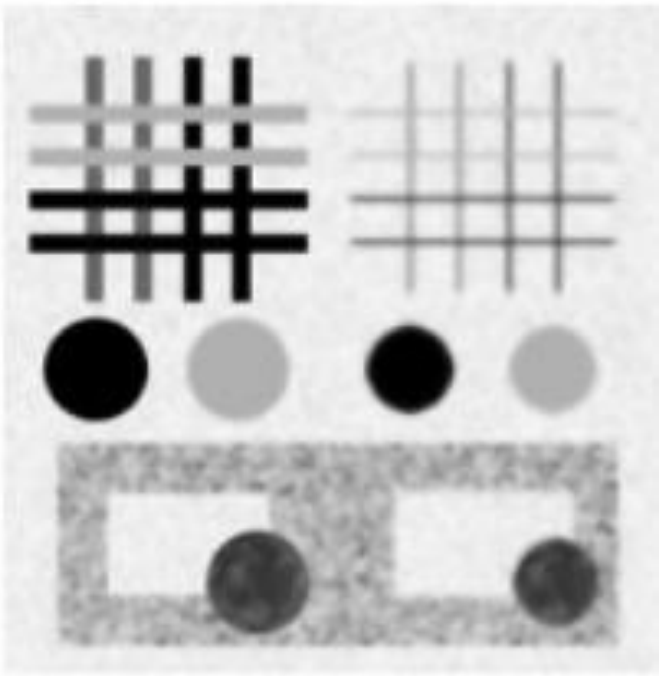


Sobel X

Source: Seiichi Uchida

Sobel Filter Y

Q: What does the above filter do?



Original

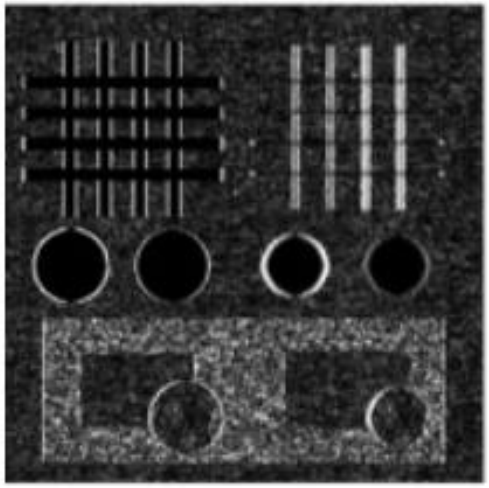
kernel

1	2	1
0	0	0
-1	-2	-1

Sobel Y

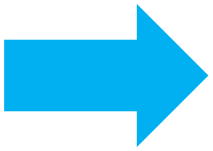
Source: Seiichi Uchida

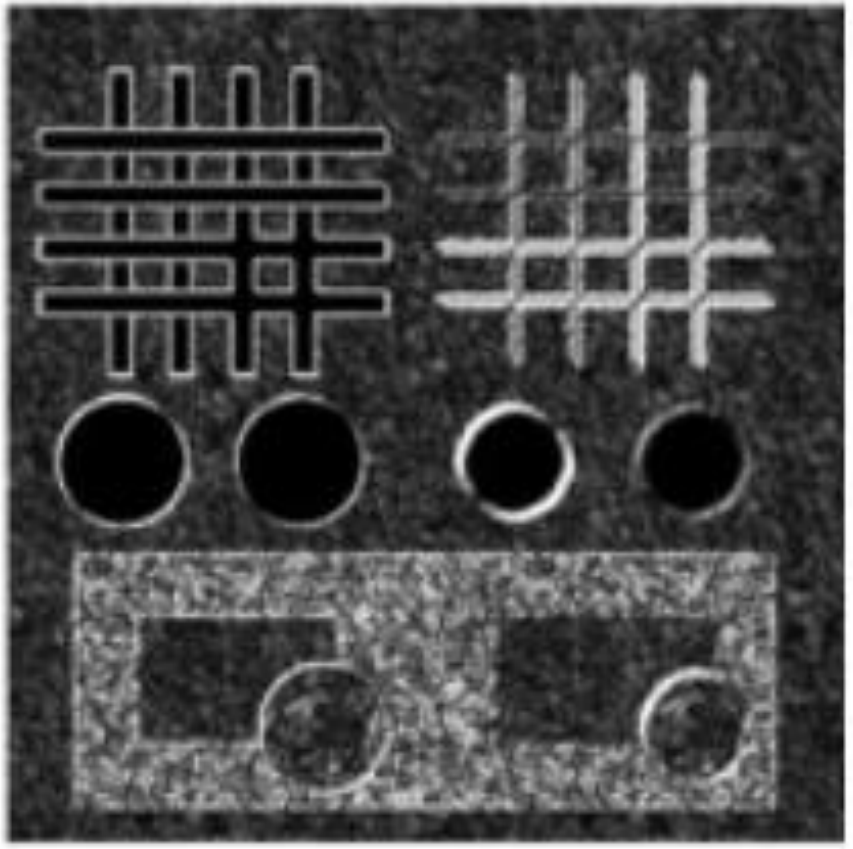
Horizontal and Vertical Combined



Sobel X

Sobel Y

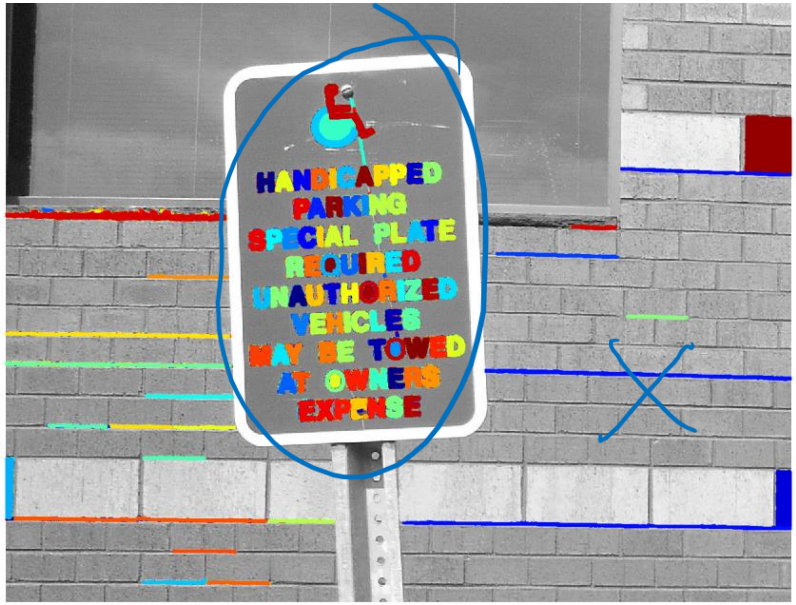




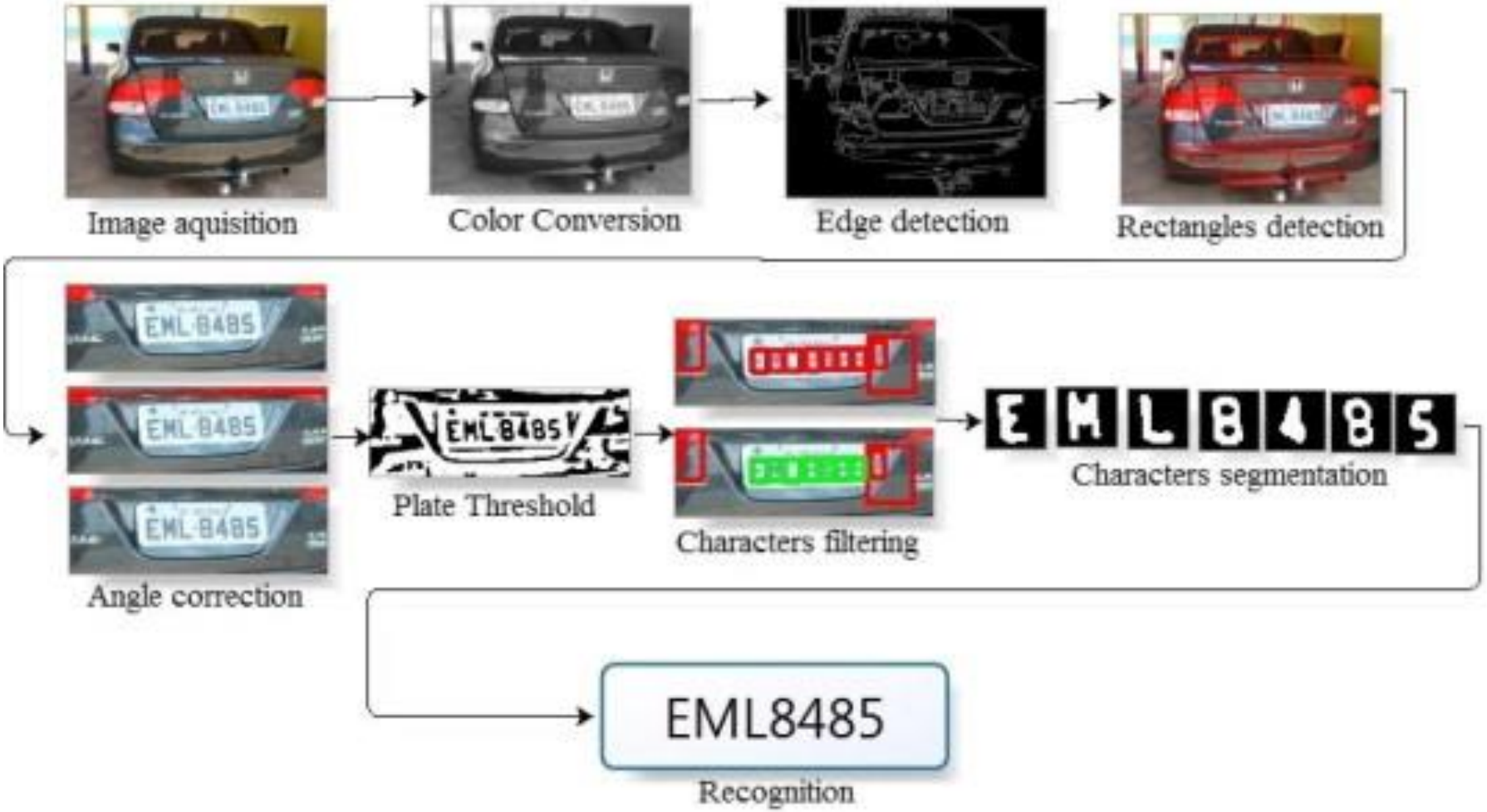
Sobel X+Y

Optical Character Recognition

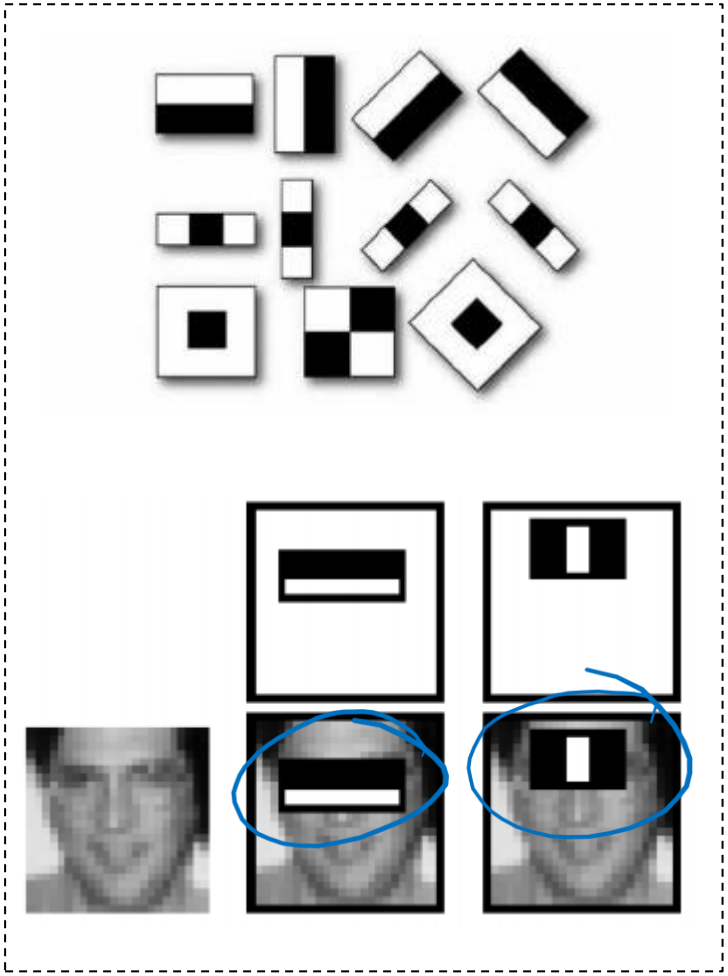
- Applies different filters and knowledge of characters to identify regions of interest
- Then eliminates regions one by one based on thresholds and line stroke characteristics



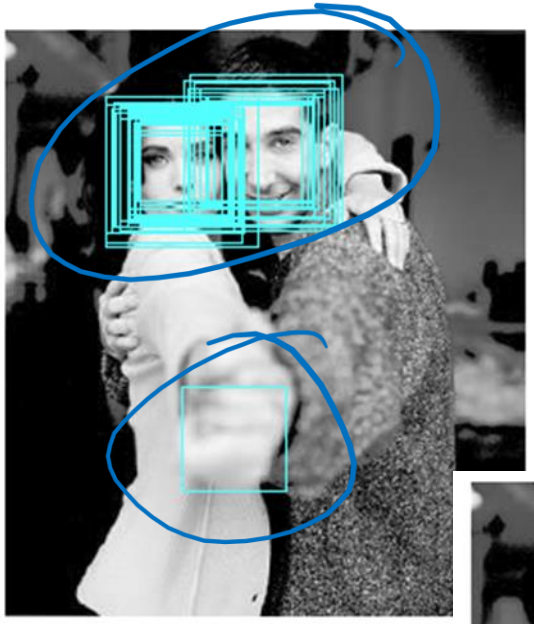
Optical Character Recognition



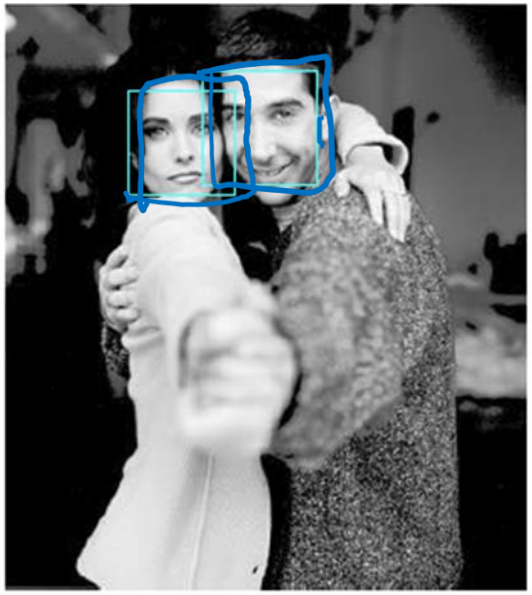
Face Detection



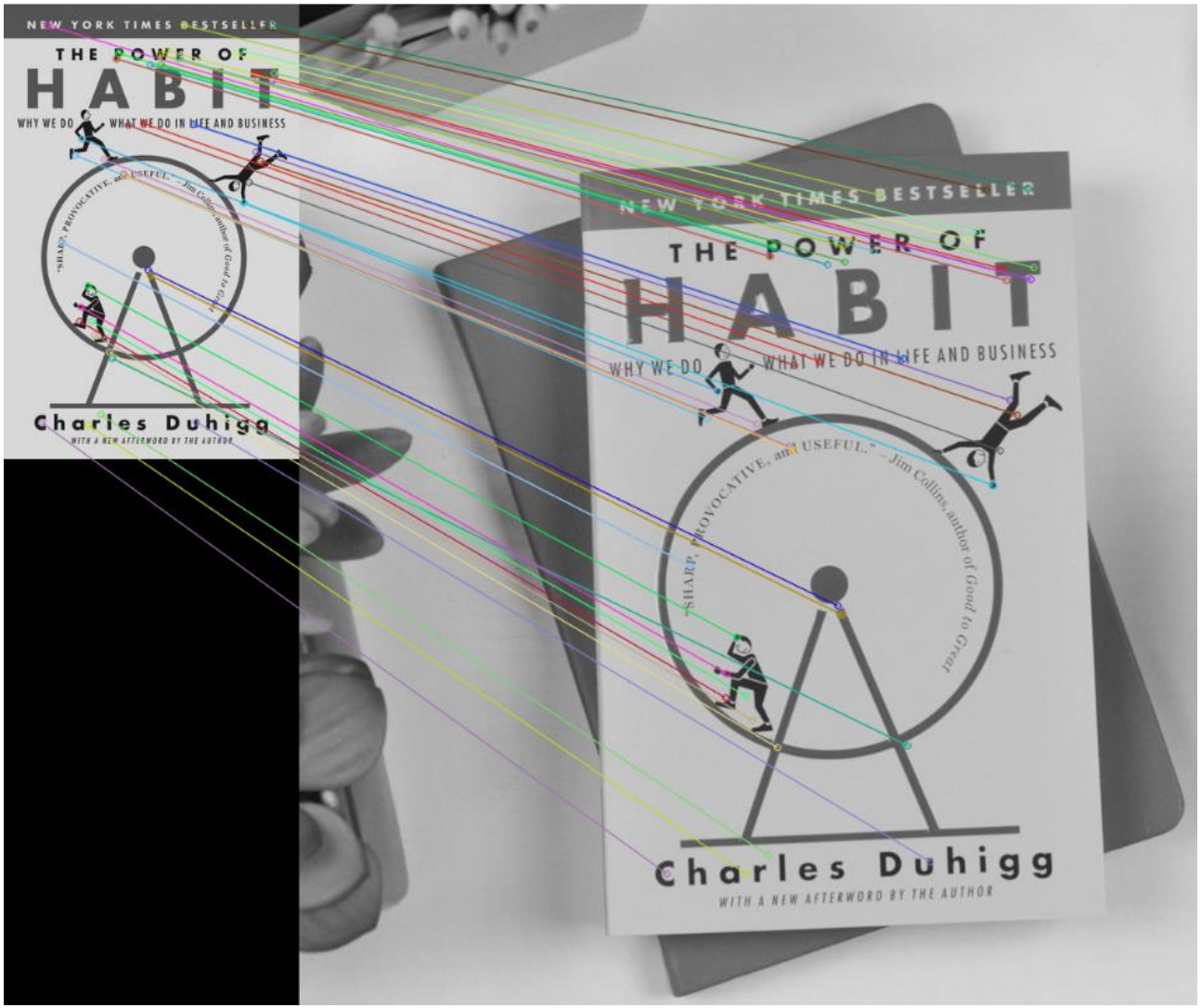
Convolutional kernels



- Viola-Jones algorithm achieved state-of-the-art results



Scale-Invariant Feature Transform (SIFT)



- Extract gradient information to find keypoints and descriptors (i.e., features) invariant to transformations
  - Translation
  - Rotation
  - Scaling

Continue at 2:13pm

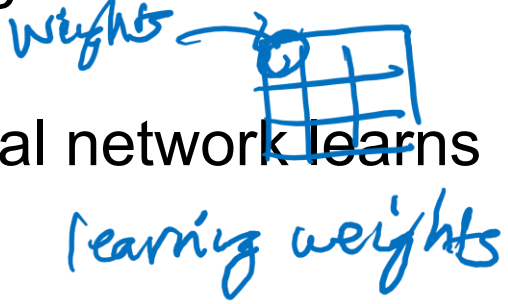
Convolutional Neural Networks

Idea:

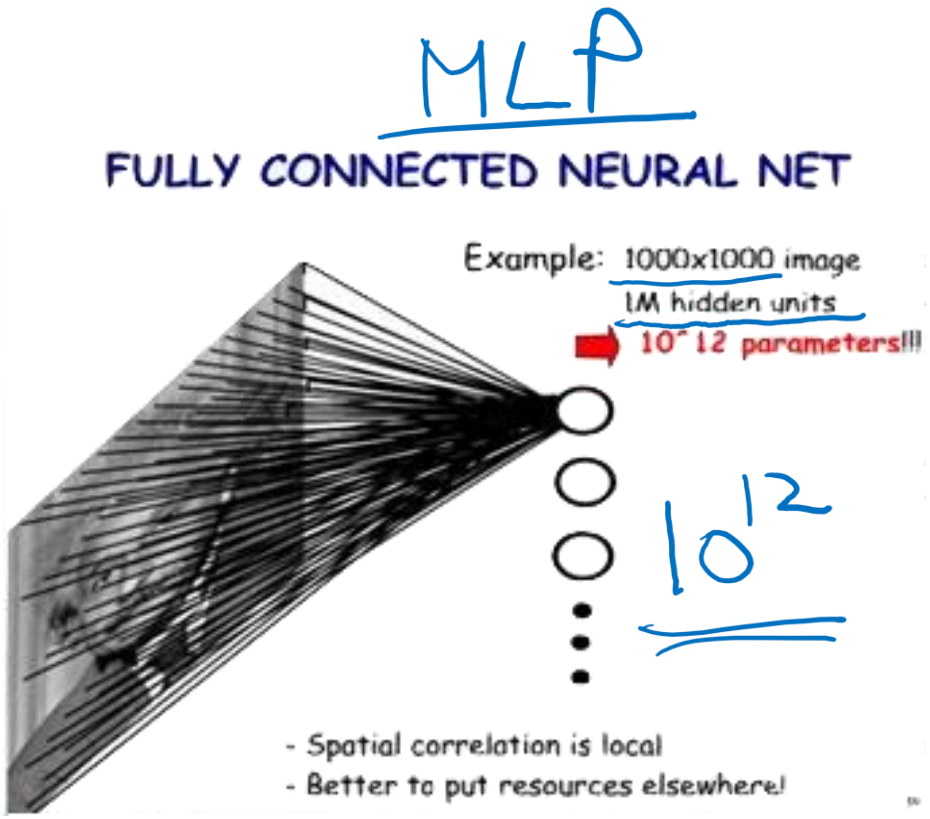
- Empower our neural networks with the ability to use convolution operations.

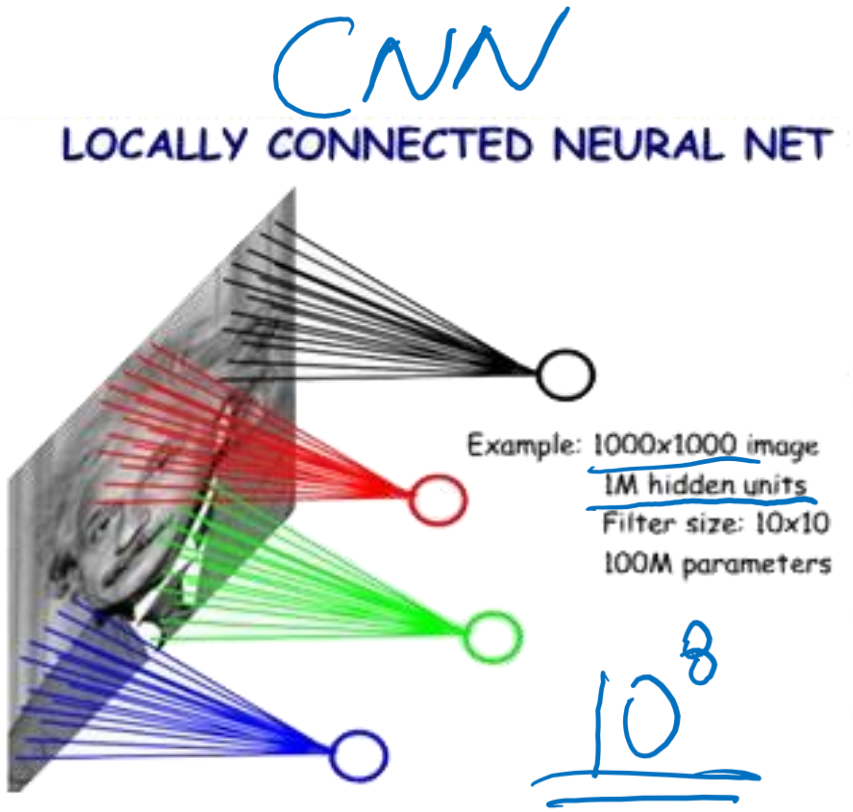
Key Points:

1. **locally-connected layers:** look for local (spatial) features in small regions of the image
2. **weight-sharing:** reuse the same weights across local regions over the entire image
3. **automate feature engineering:** neural network learns the kernel values, which become our parameters/weights



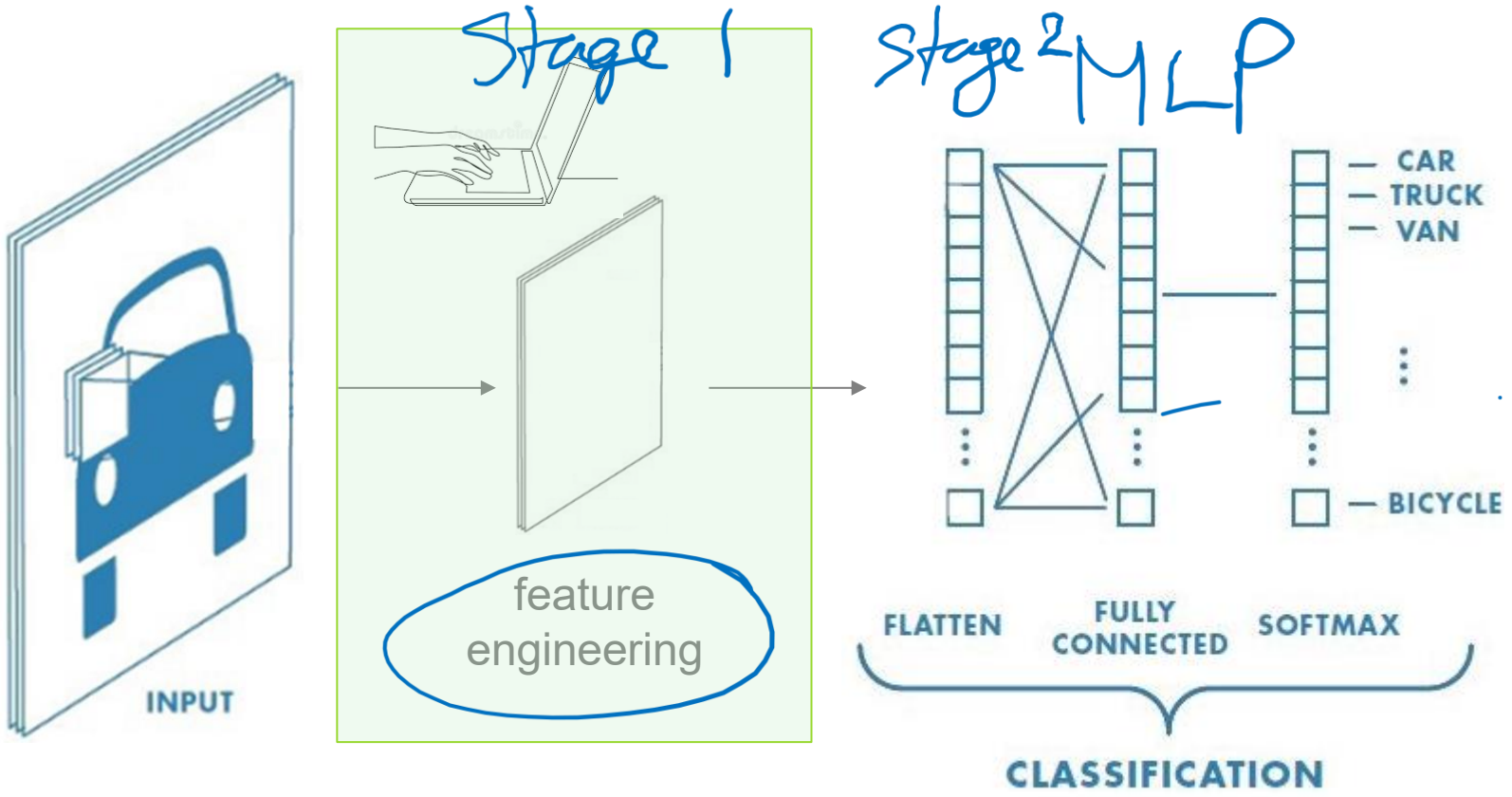
Requires Less Parameters





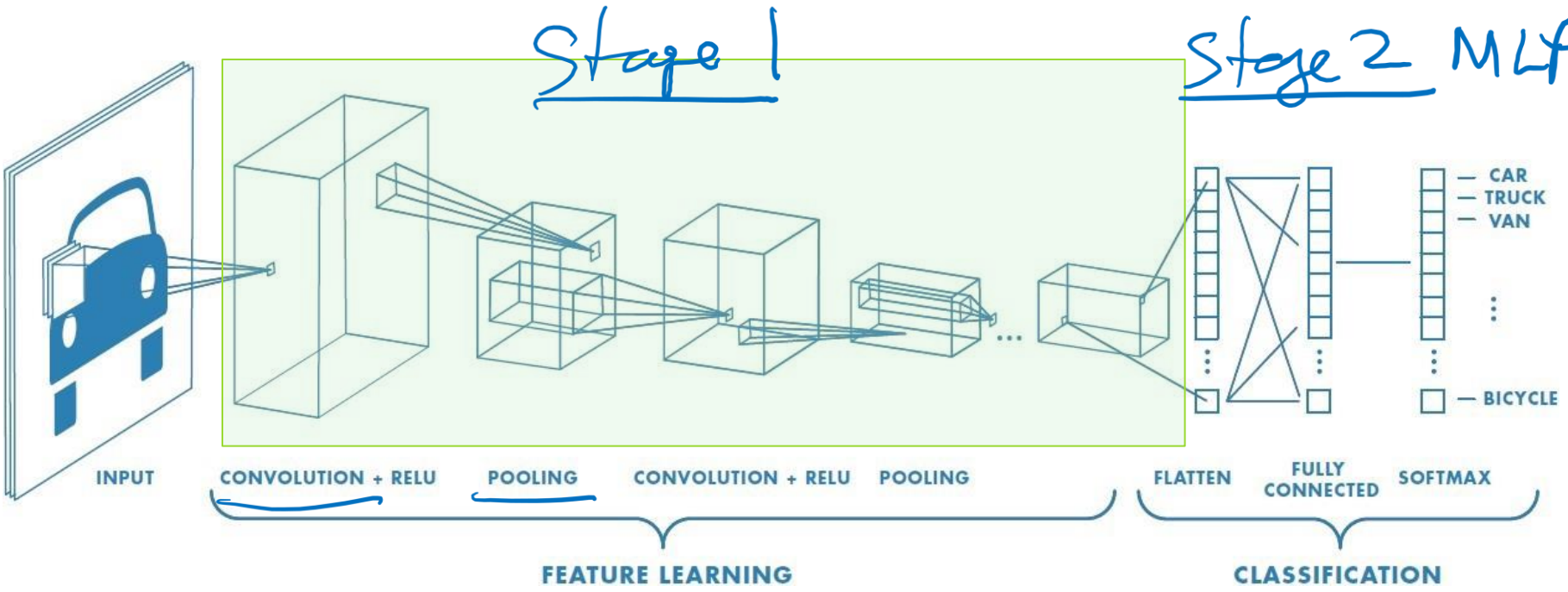
Convolutional Layers

Traditional Machine Learning



Most of the effort went into feature engineering or processing of the input.

Deep Learning



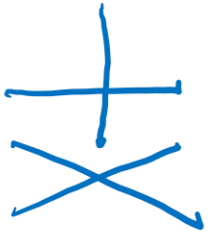
Automate the feature engineering process

Architecture **inspired by the organization of neurons in the visual cortex**

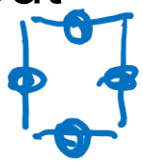
Biological Influence

Hubel and Wiesel Cat Experiments (1958 – 1959):

- Individual neurons respond to stimuli only in a restricted region of the visual field known as the **Receptive Field**.
  - collection of such fields overlap to cover the entire visual area.
  - some neurons react only to images of horizontal lines, while others react to line orientations



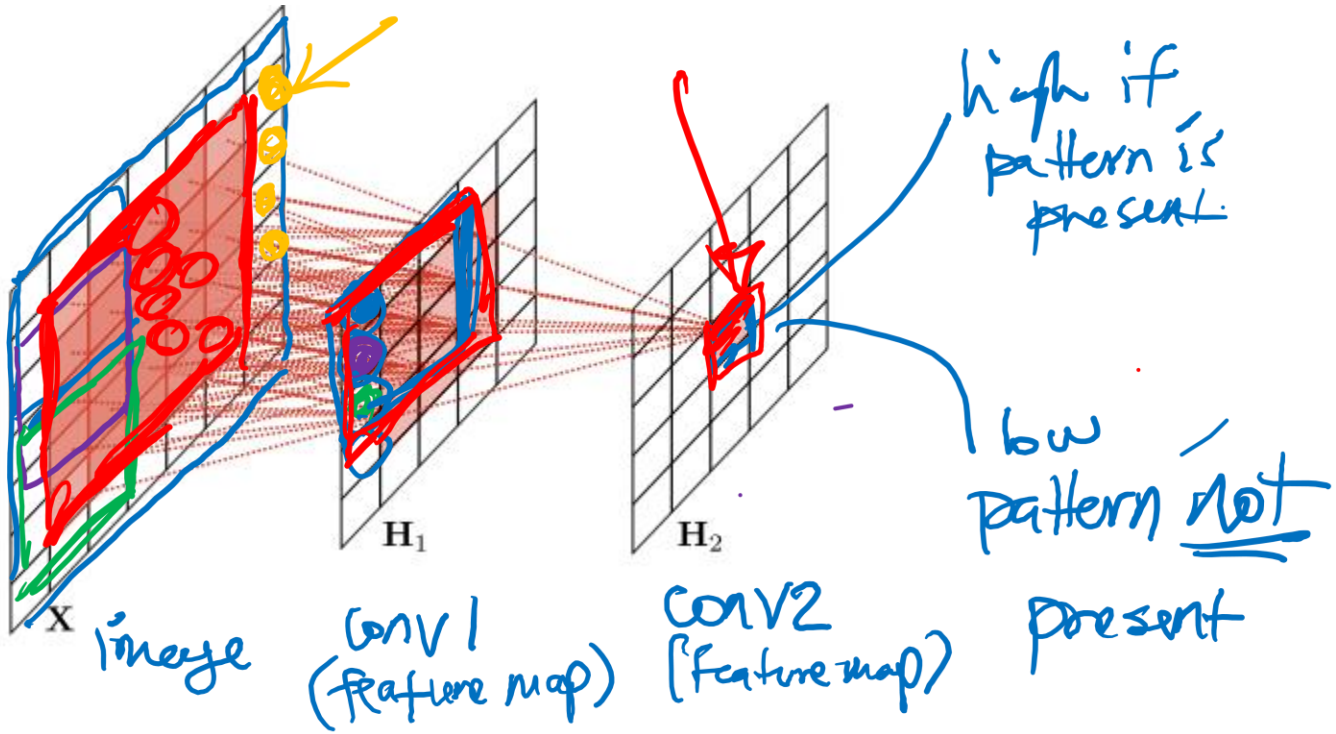
  - two neurons may have the same receptive field but react differently



  - higher-level neuron inputs are the outputs of neighbouring lower-level neurons

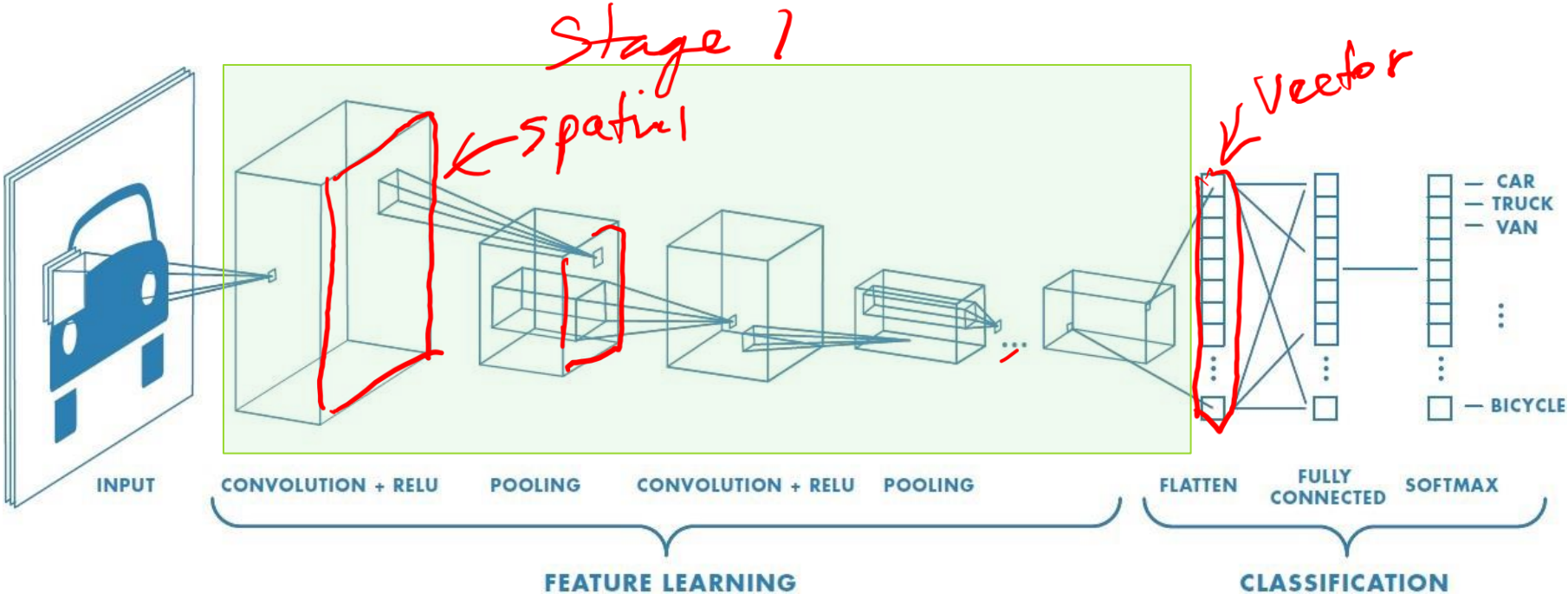
Receptive Fields

Refers to all the pixels in the input which influence the neuron being examined.



example receptive field of neuron H_2

Architecture: CNN Layers



Introduce two new types of layers:

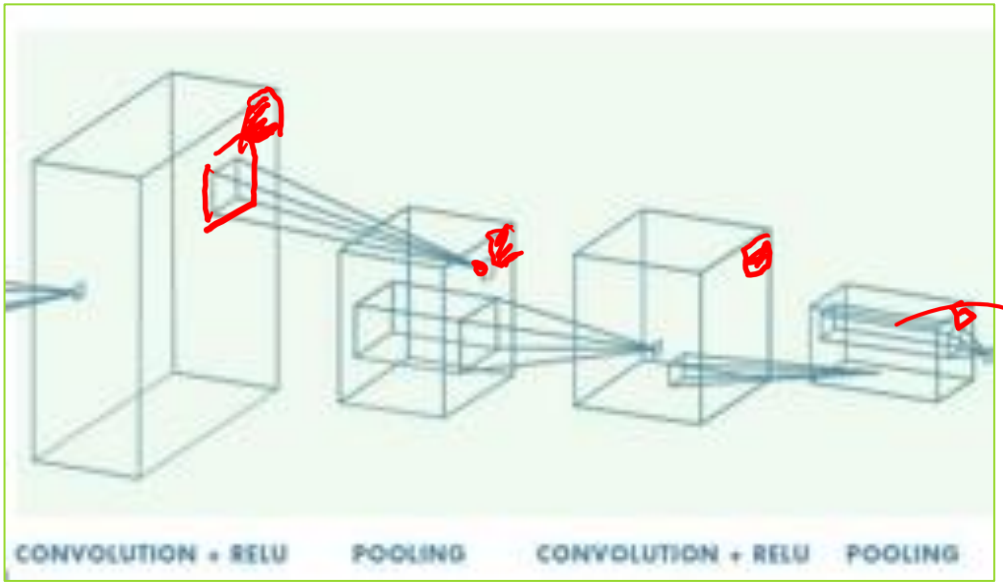
1. Convolutional layers
2. Pooling layers

important
optional

Q: What is the geometry of the convolution and pooling layers?

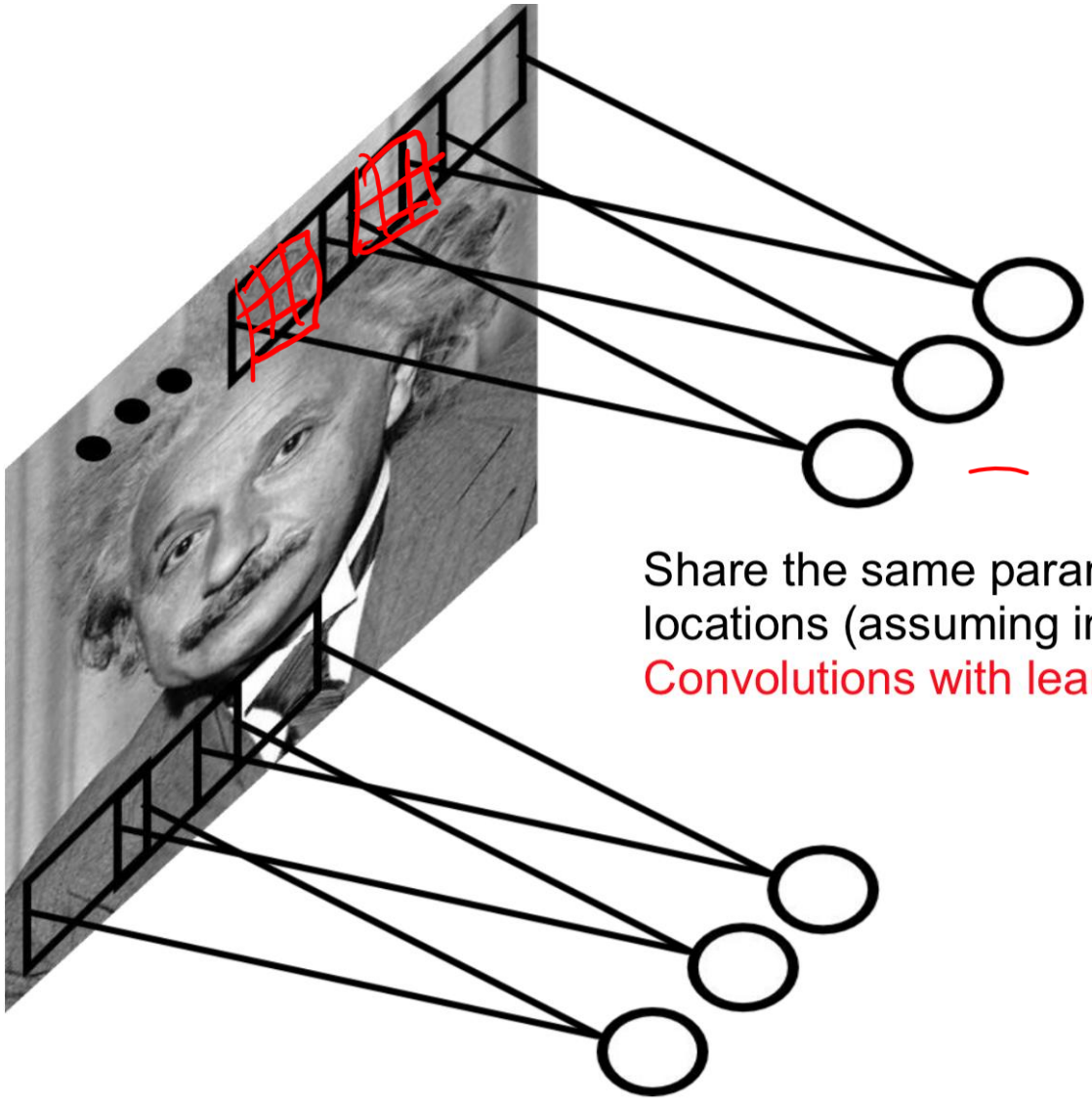
Locally Connected Layers

- **Notion of proximity:** To find out what object is at a particular pixel, can look at other pixels nearby.
  - nearby pixels will be more informative than pixels further away



maintains geometry (top-right corner of the hidden layers being computed from the top-right region of the original input image)

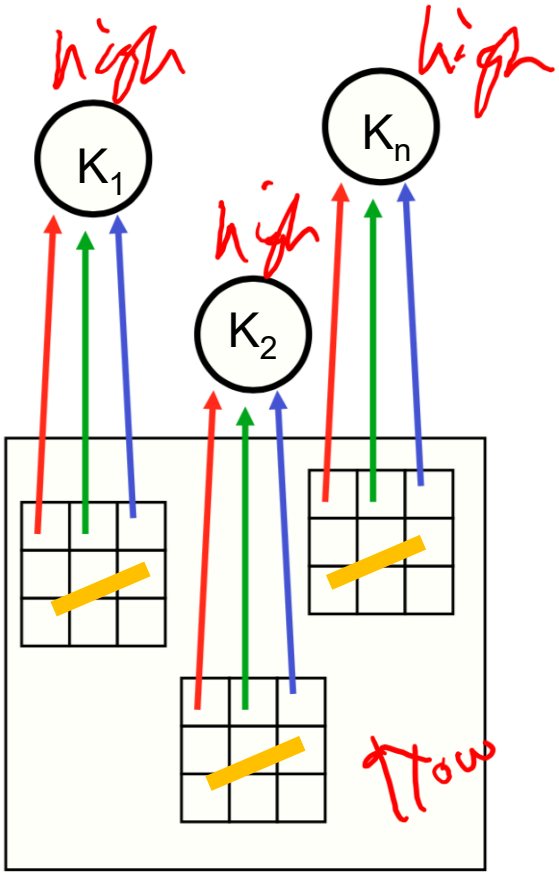
Weight Sharing



Share the same parameters across different locations (assuming input is stationary):
Convolutions with learned kernels

Weight Sharing

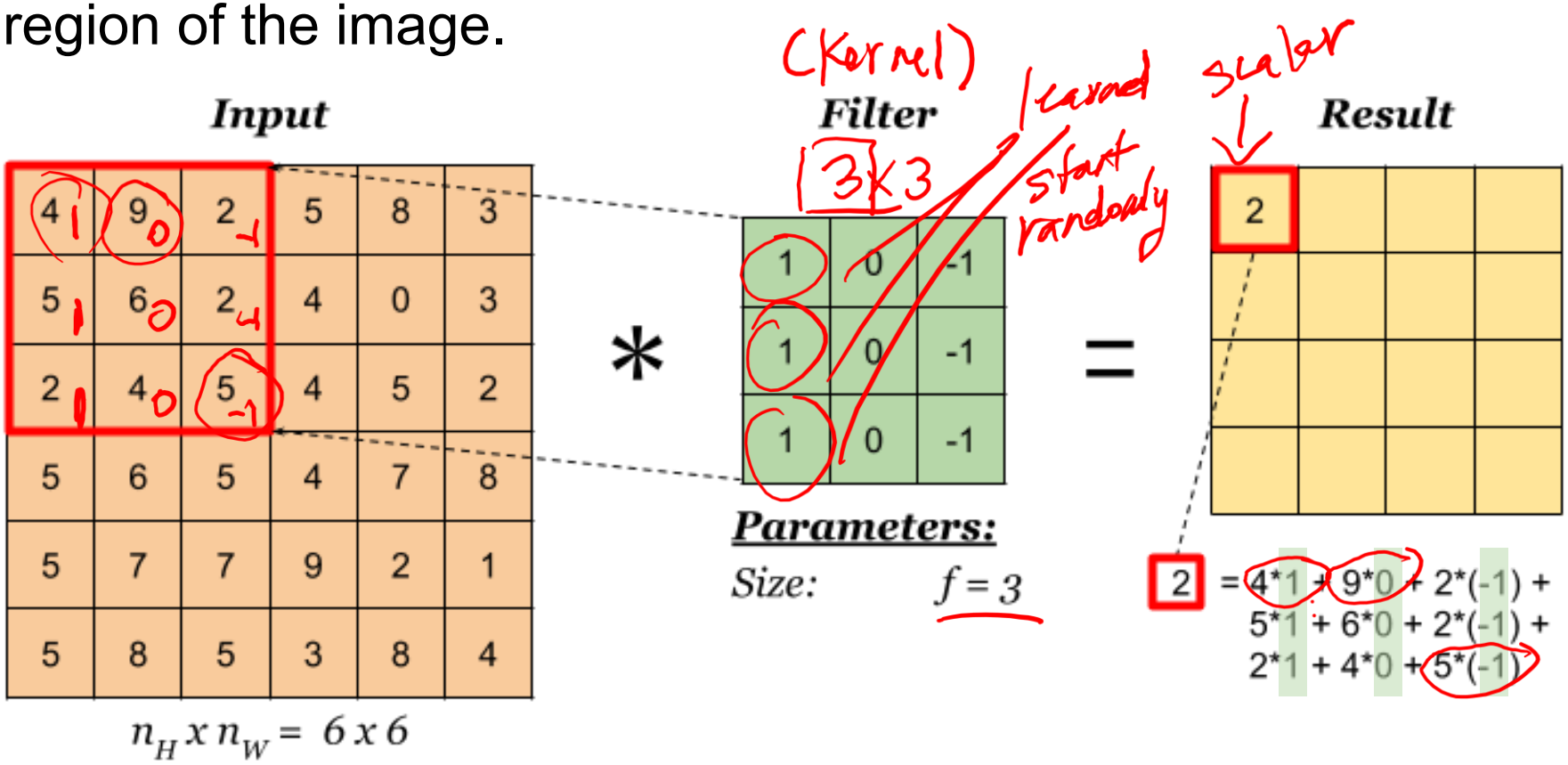
Each neuron on the higher layer is detecting the same feature, but in different locations on the lower layer



- "Detecting" = the output (activation) is high if the feature is present
- "Feature" = something in the image, like an edge, blob or shape

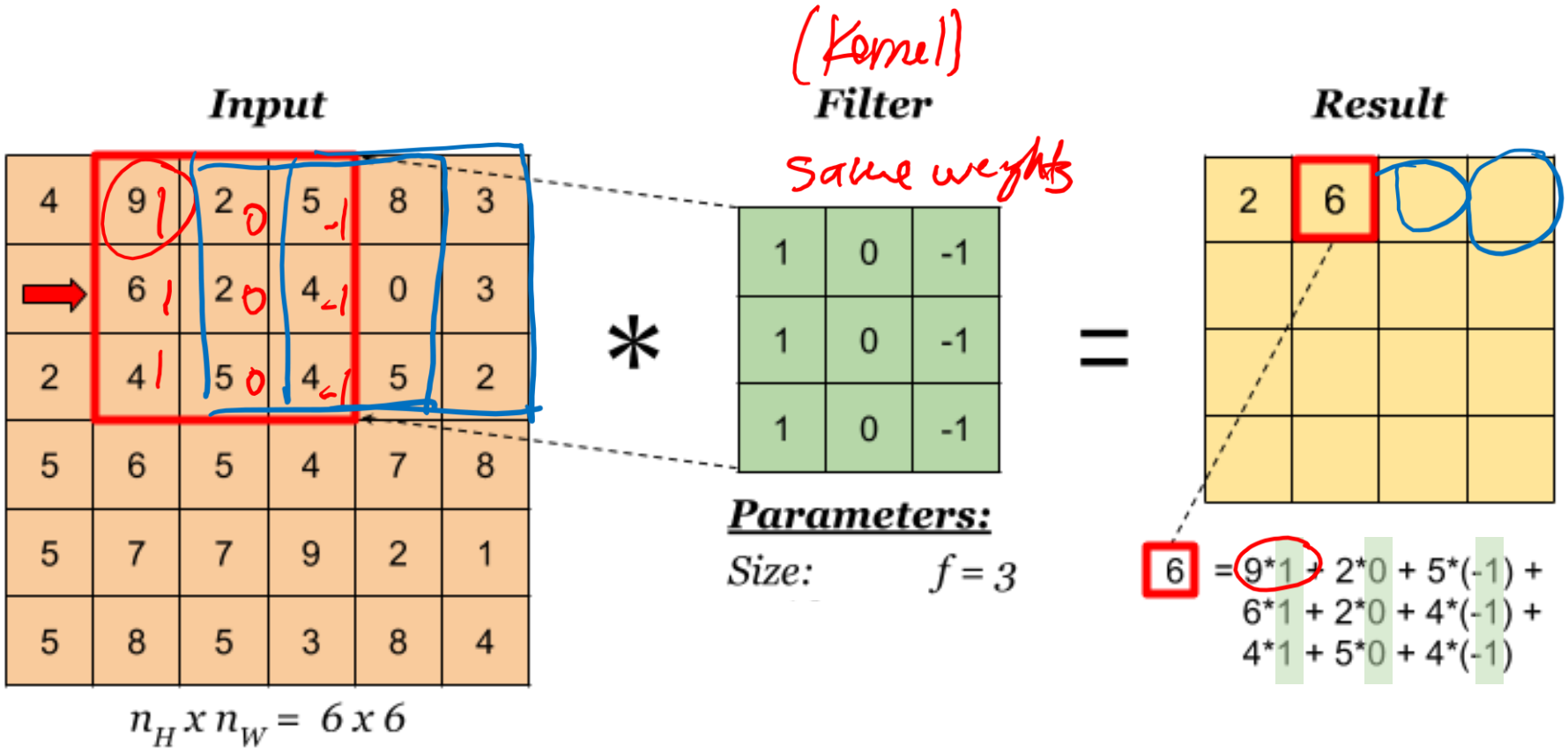
Convolution Layer Forward Pass

To compute the output, we superimpose the kernel on a region of the image.

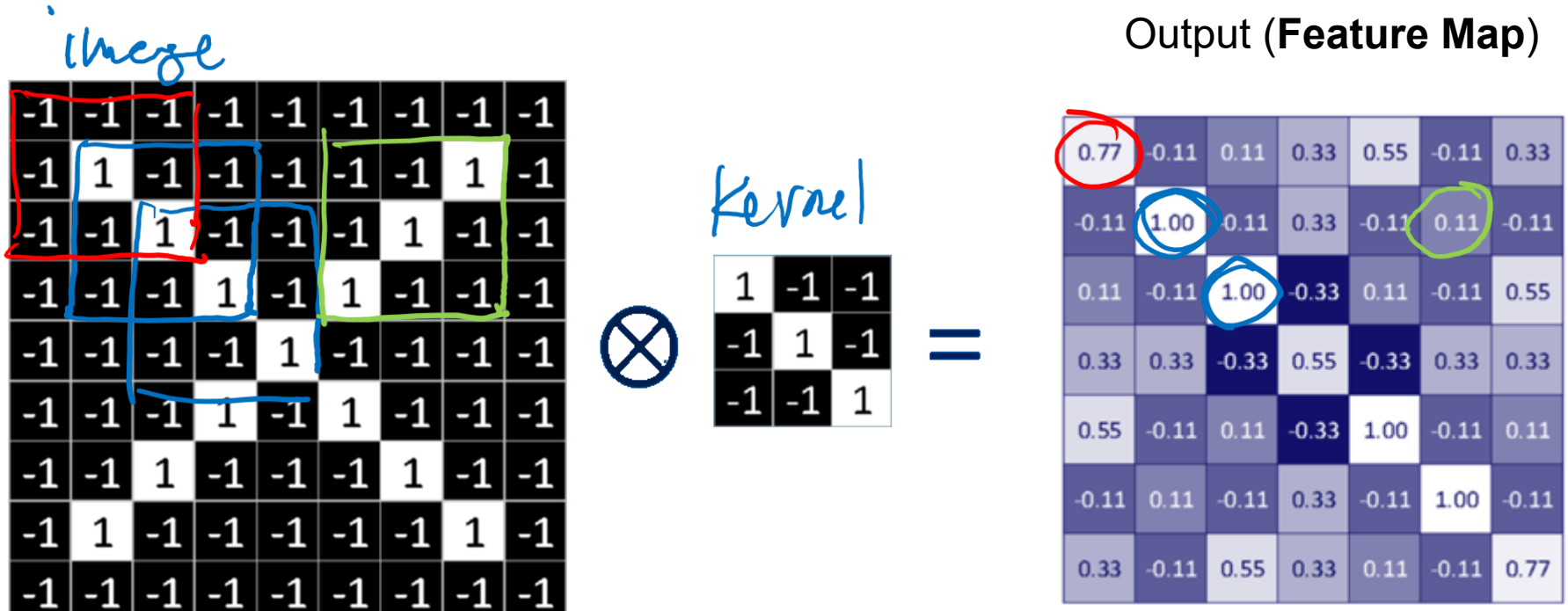


Convolution Layer Forward Pass Con't

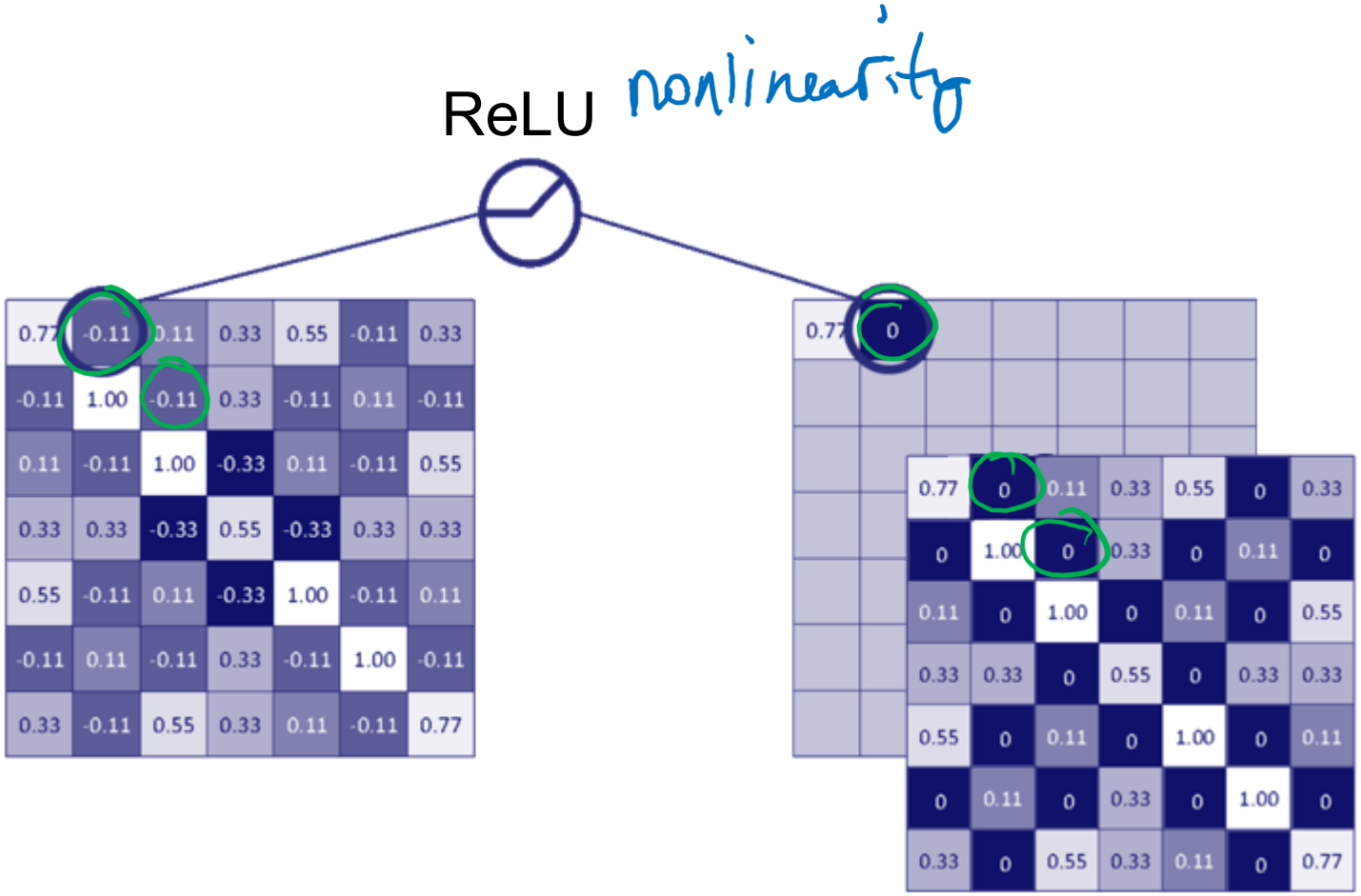
Slide one to the right and compute



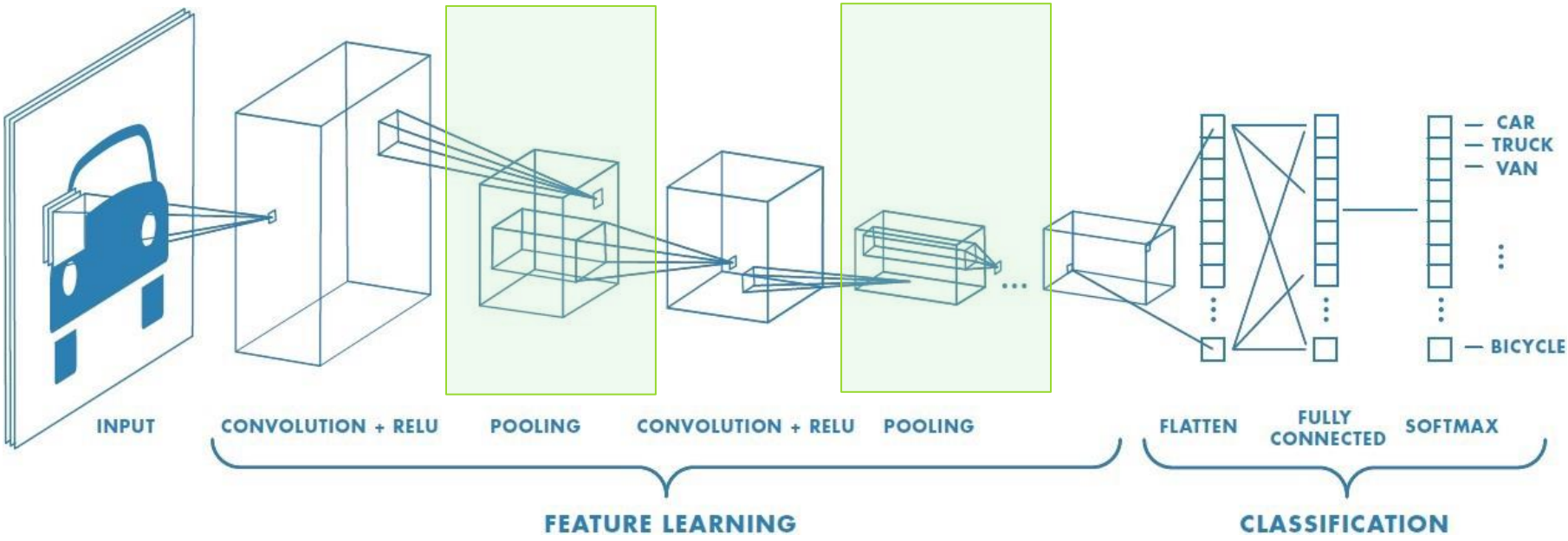
Example Calculation - Feature Map



Example Calculation - Activation



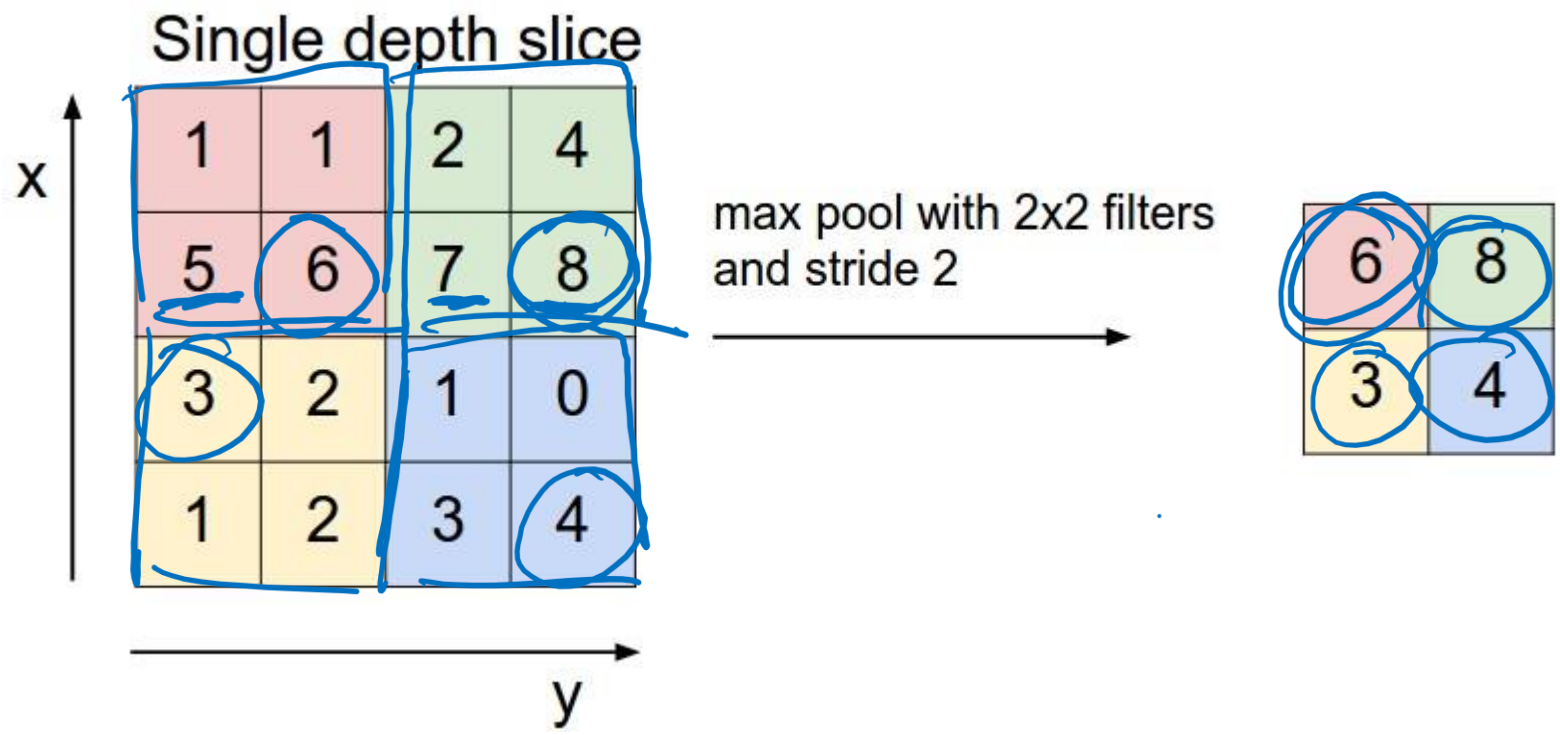
Pooling Layers



Pooling is a technique used to consolidate information.

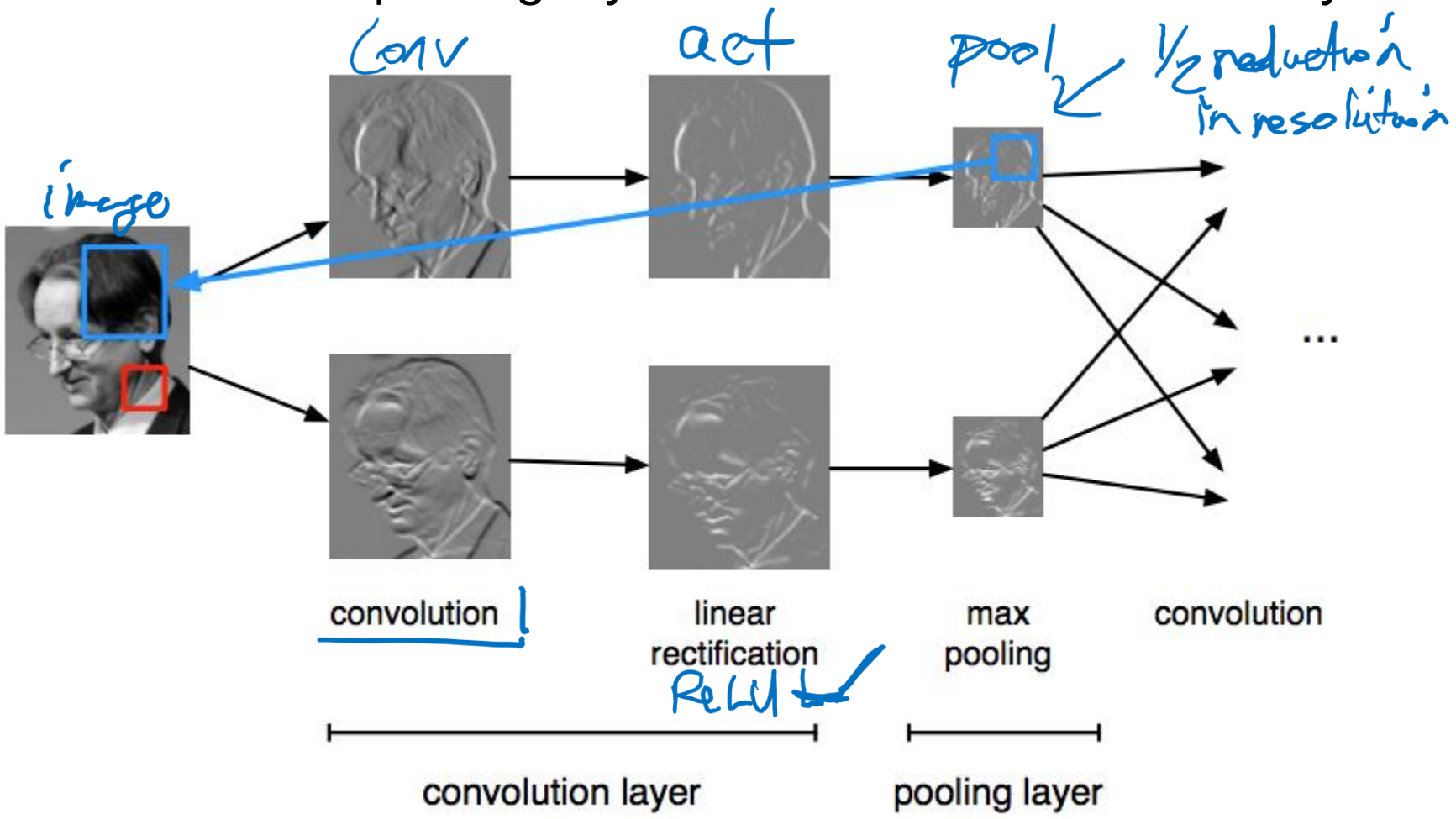
Max-Pooling

Idea: take the maximum value in each 2 × 2 grid



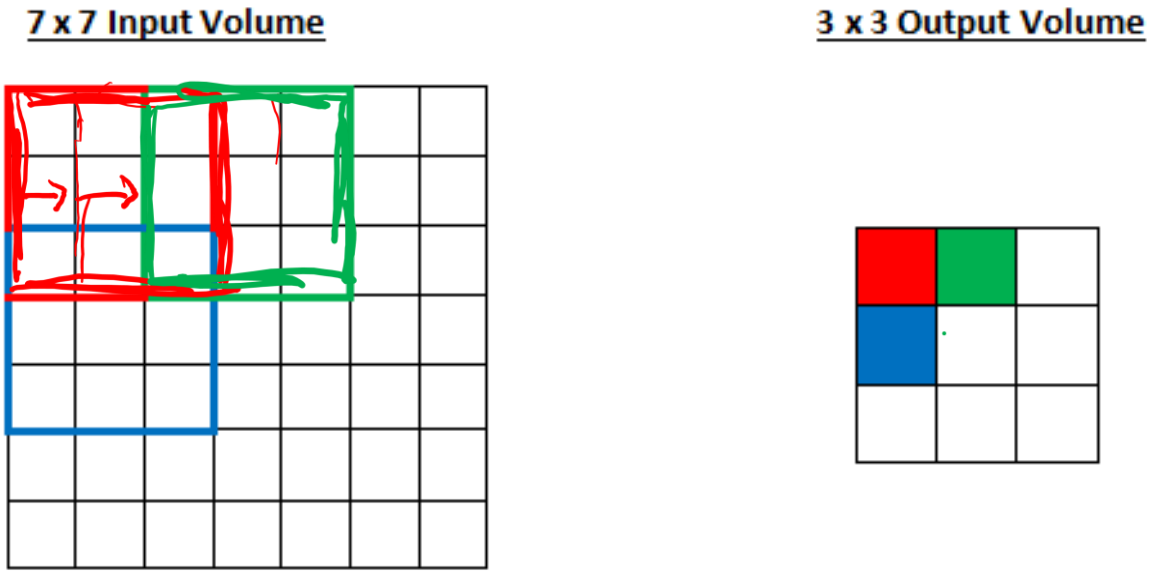
Max-Pooling Example

We can add a max-pooling layer after each convolutional layer



Strided Convolution

- Pooling operations less common now and generally replaced with strided convolutions



Shift the kernel by 2 (stride=2) when computing the next output feature.

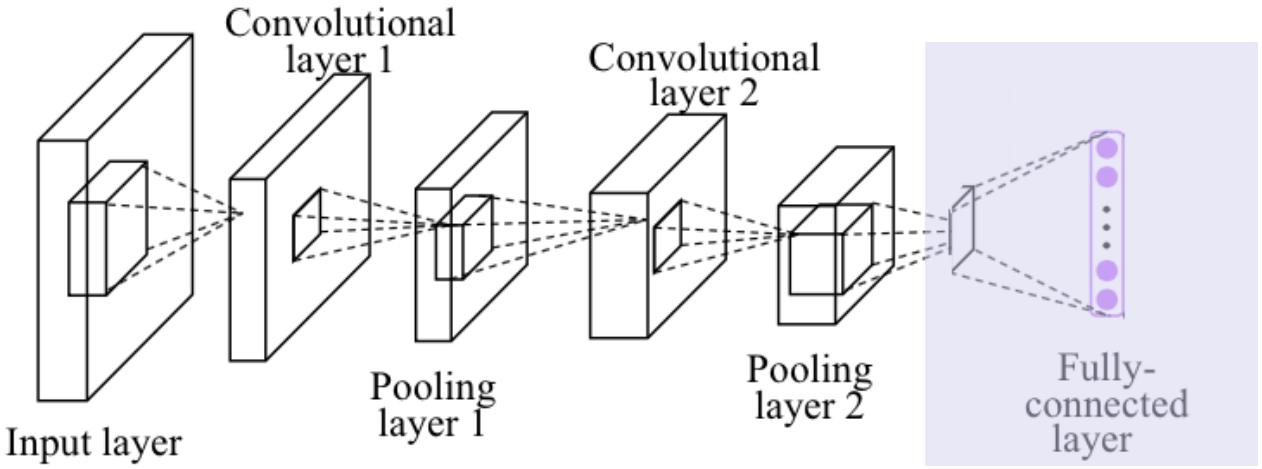
Continue at 3:01pm

PyTorch Implementation

CNNs in PyTorch: Linear Layers

Let us make a simple CNN in PyTorch

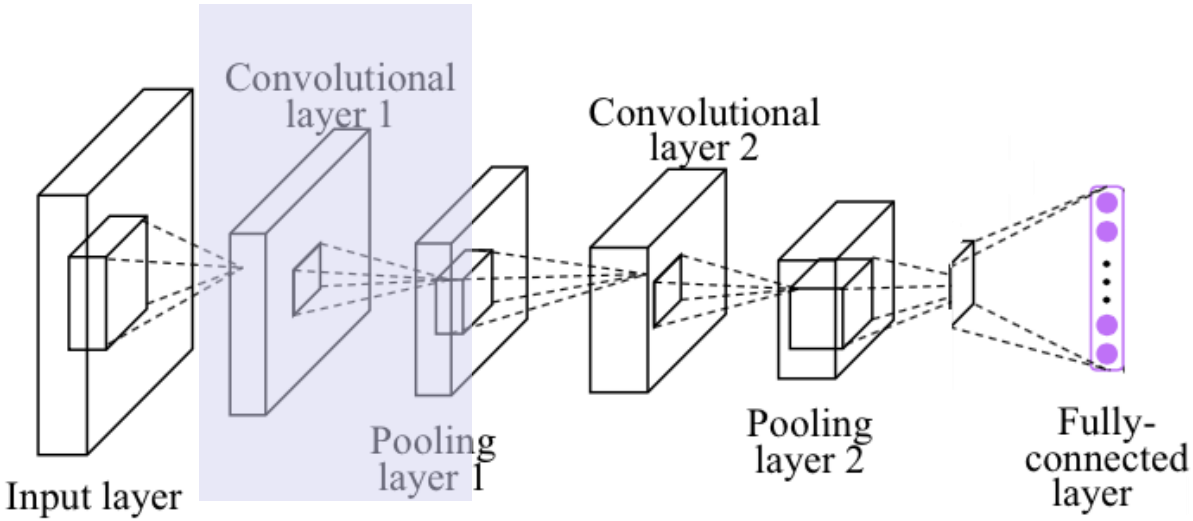
- Recall that in PyTorch, we can create a fully-connected linear layer between successive layers like this




```
self.fc = nn.Linear(n, 10)
```

CNNs in PyTorch: Convolutional Layers

In PyTorch, we can create a convolutional layer using `nn.Conv2d`:



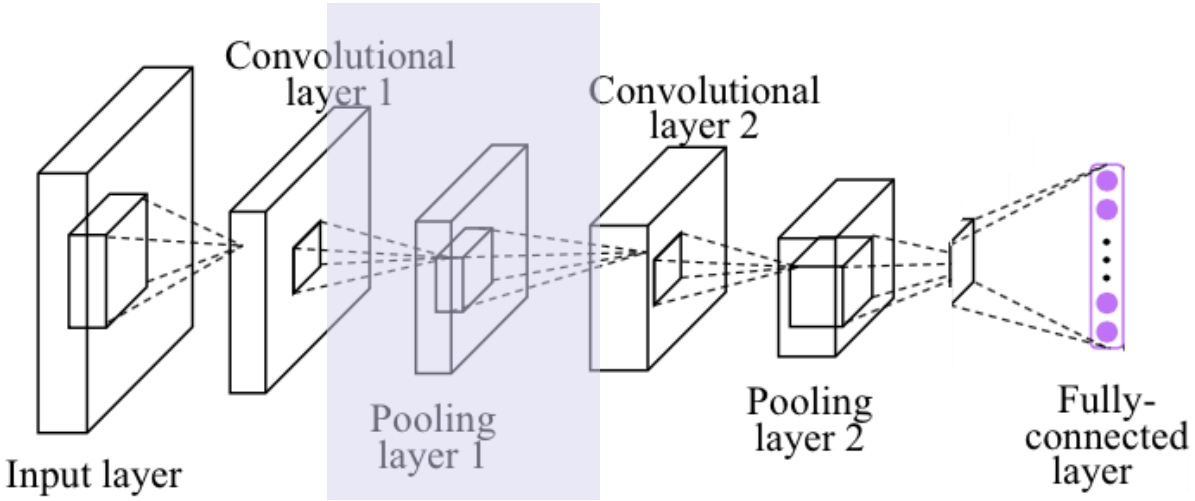

```
self.conv1 = nn.Conv2d(in_channels = 3,
                       out_channels = 7,
                       kernel_size= 5,
                       stride = 1,
                       padding = 1)
```

likely an RGB

default stride 1 pad 0

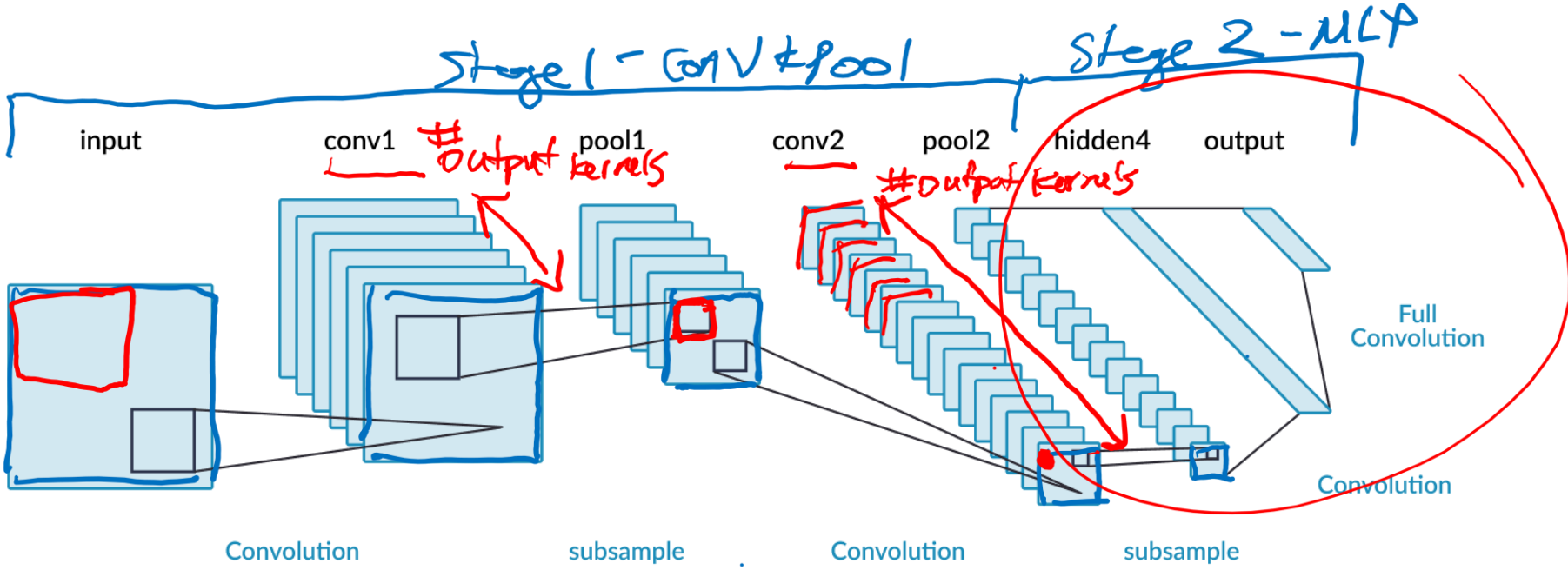
CNNs in PyTorch: Max Pooling Layers

In PyTorch, we can create a max pooling layer using `nn.MaxPool2d`:




```
self.pool = nn.MaxPool2d(kernel_size = 2,
                         stride = 2)
```

CNN Architecture



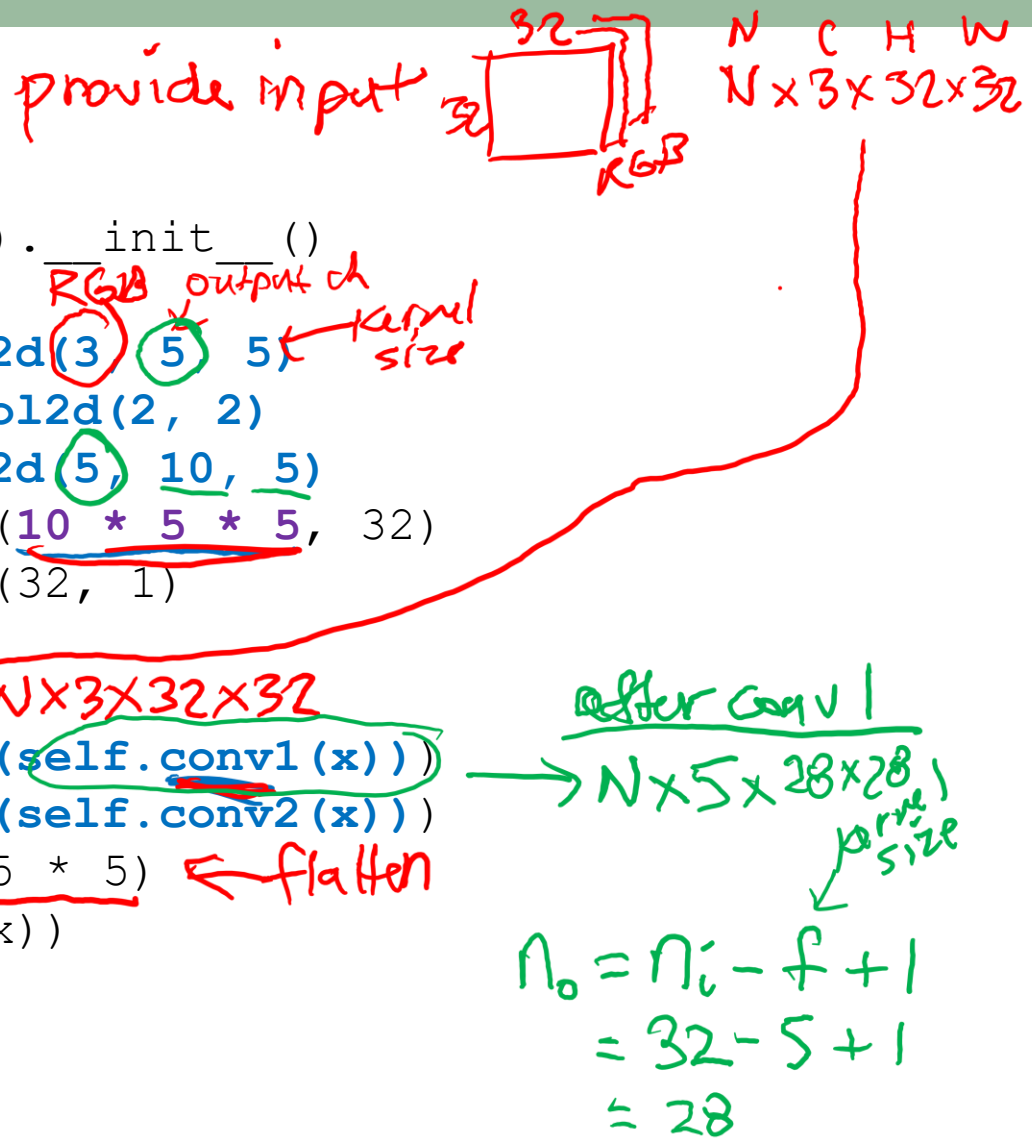
1. H x W dim of feature maps reduced

Consolidation of information!

2. # of output channels (kernels) increases

PyTorch Example CNN Architecture:

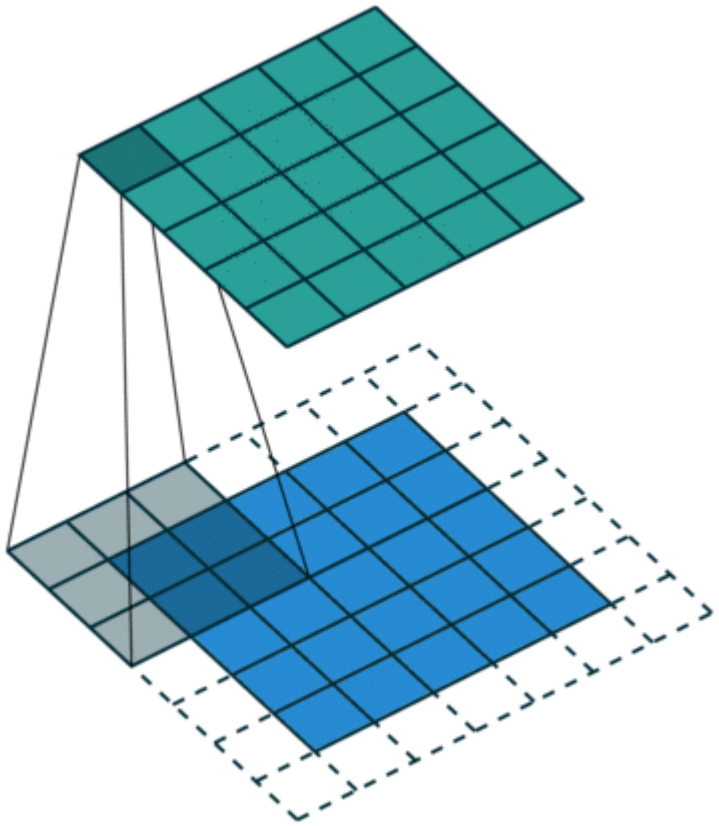
```
class LargeNet(nn.Module):
    def __init__(self):
        super(LargeNet, self).__init__()
        self.name = "large"
        self.conv1 = nn.Conv2d(3, 5, 5)
        self.pool = nn.MaxPool2d(2, 2)
        self.conv2 = nn.Conv2d(5, 10, 5)
        self.fc1 = nn.Linear(10 * 5 * 5, 32)
        self.fc2 = nn.Linear(32, 1)

    def forward(self, x):
        x = self.pool(F.relu(self.conv1(x)))
        x = self.pool(F.relu(self.conv2(x)))
        x = x.view(-1, 10 * 5 * 5)
        x = F.relu(self.fc1(x))
        x = self.fc2(x)
        return x
```



PyTorch Examples

More on CNNs

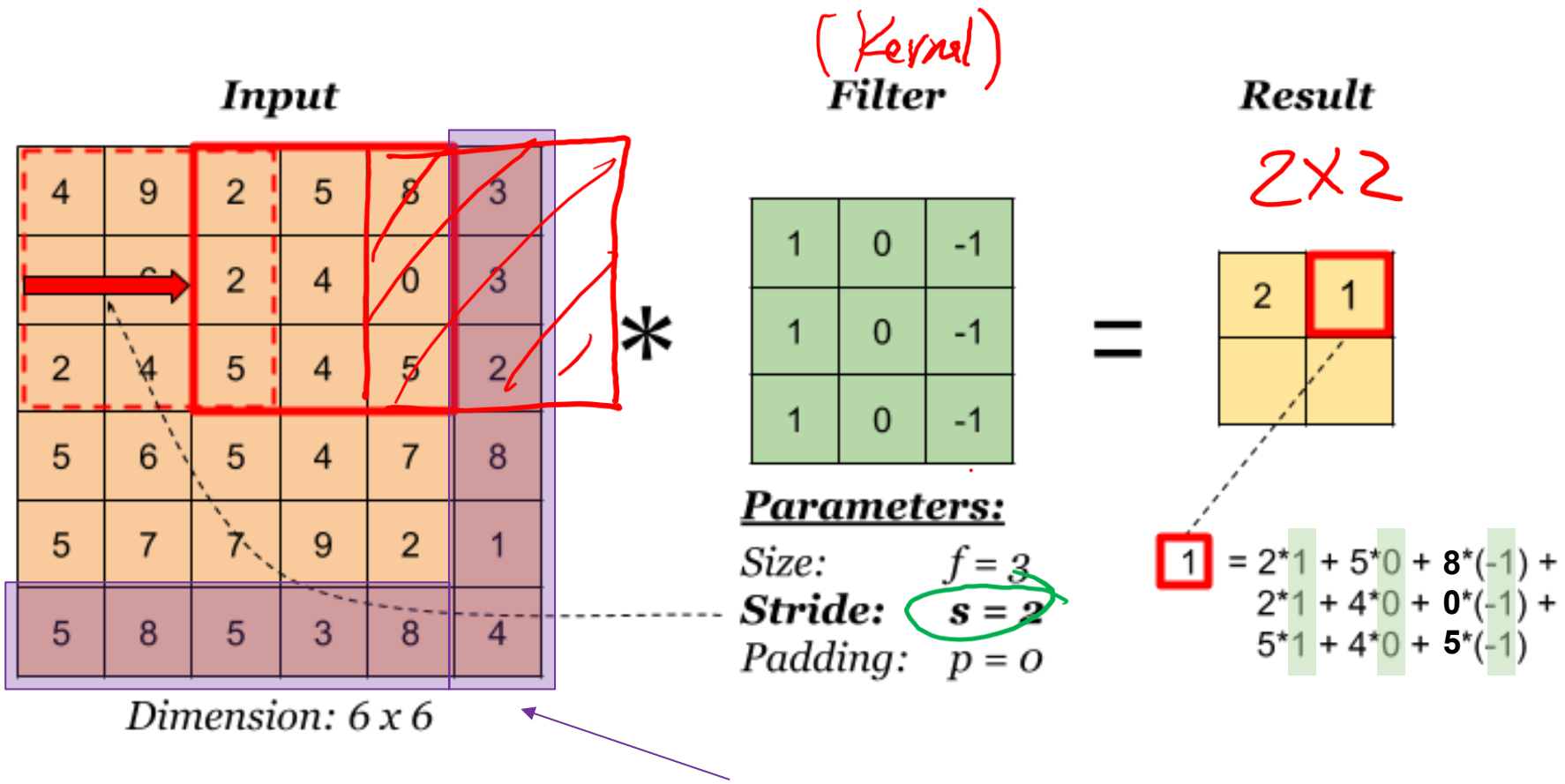
Zero Padding



Can add zeros around the border of the image before convolution

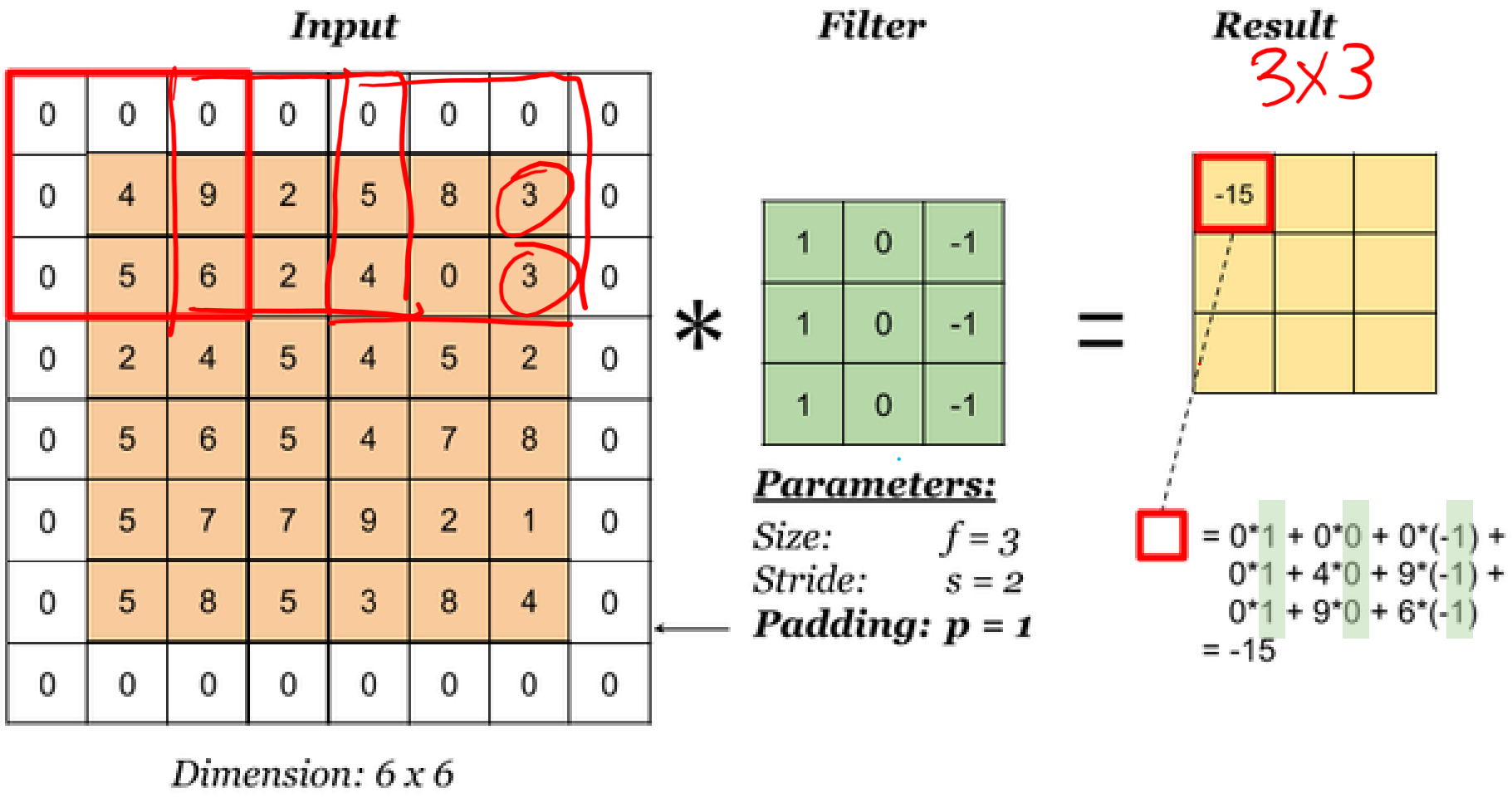
Q: Why would we want to add zero padding?

Convolution Calculation: Stride

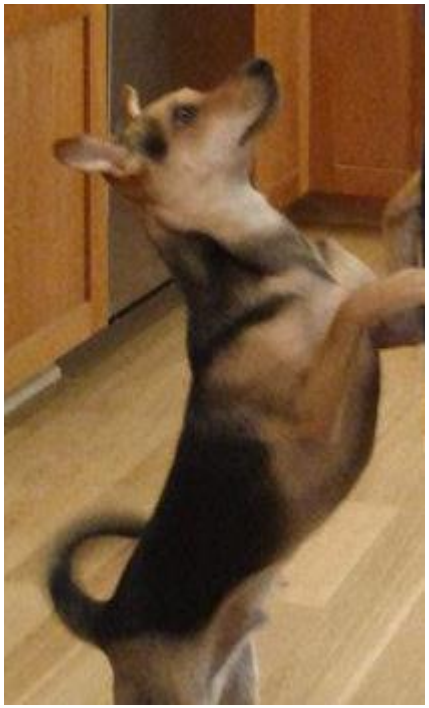


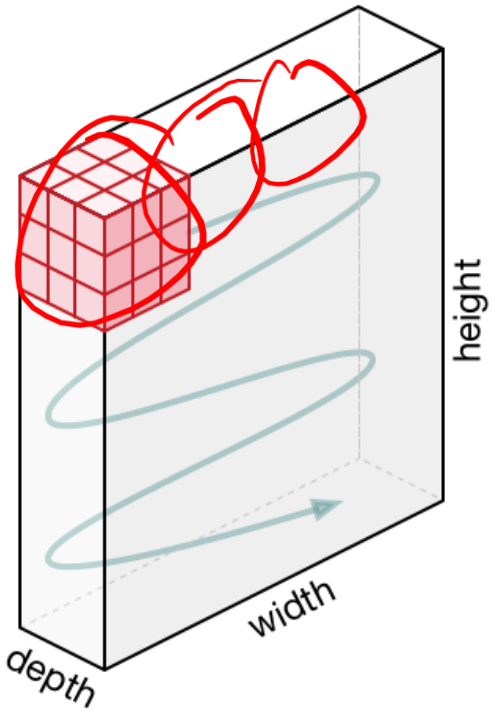
Edges not considered in the computations due to Stride = 2

Convolution Calculation: Padding

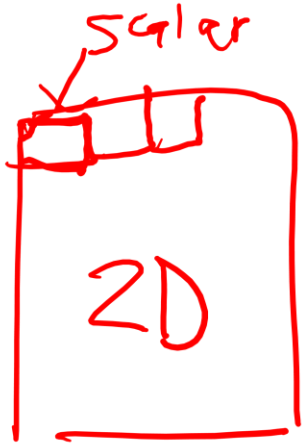


What if we have 3 colours?





Convolution in RGB

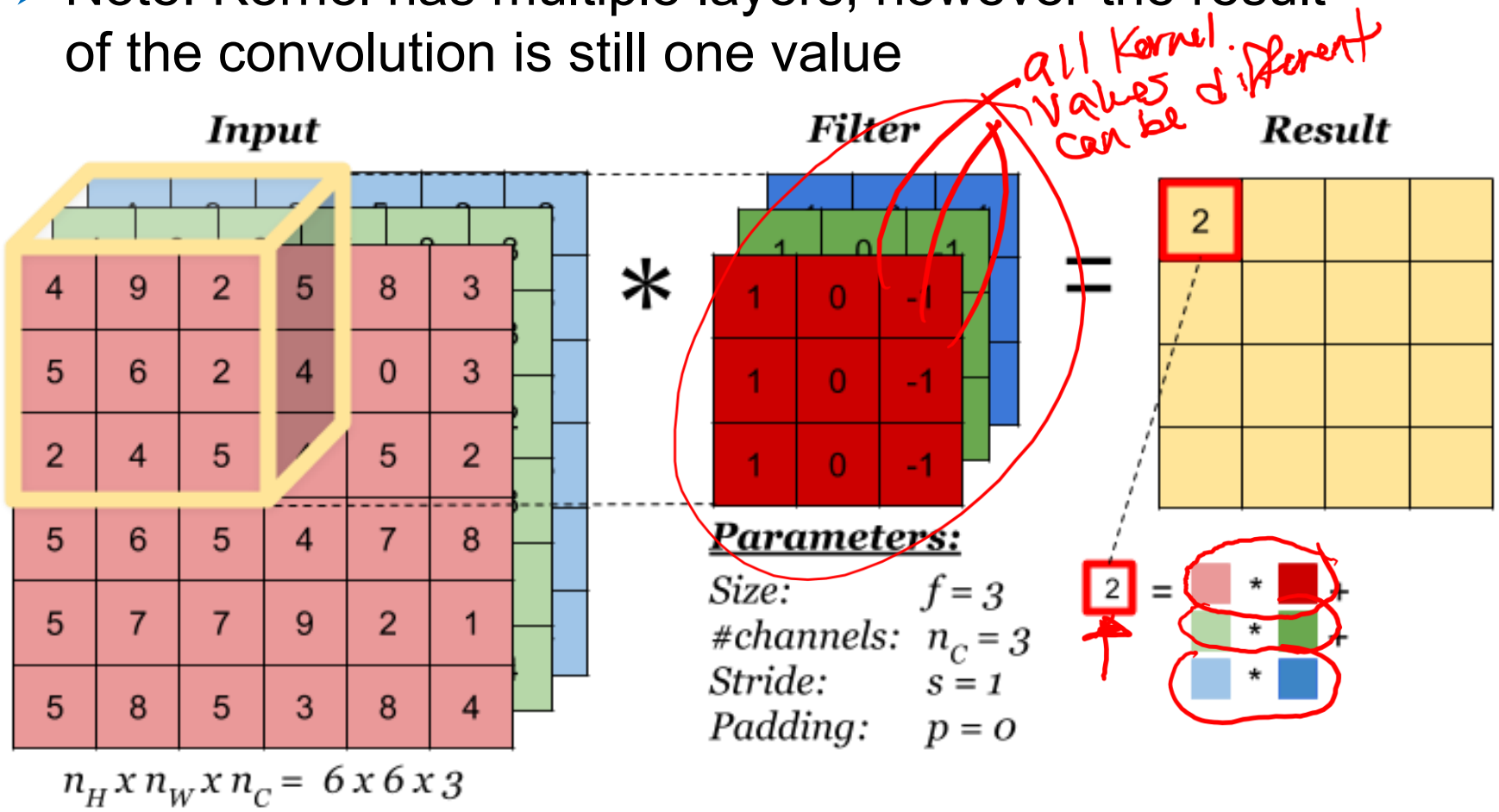


The kernel becomes a 3-dimensional tensor!

In this example, the kernel has size 3 ×3 × 3

RGB Convolution

- Note: Kernel has multiple layers, however the result of the convolution is still one value



Convolutions: RGB Input

Colour input image: 3 × 28 × 28

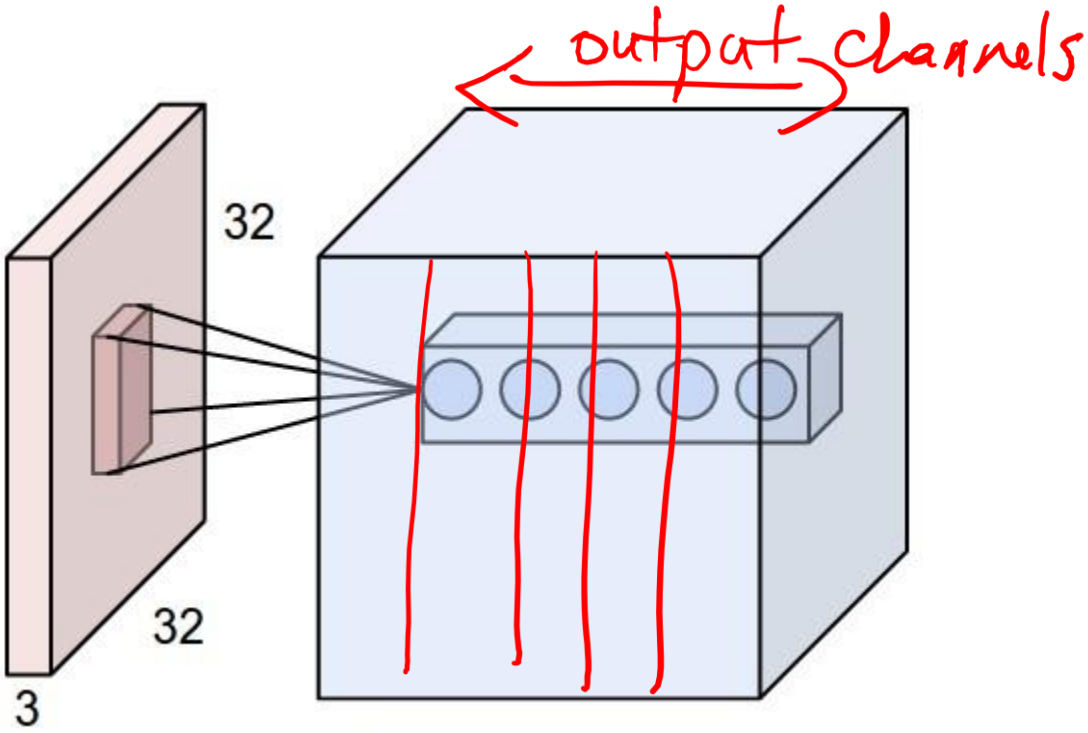
Convolution kernel: 3 ×3 × 3

Questions:

- How many feature maps are in the output layer (higher layer)? What are the dimensions?
- How many trainable weights are there?

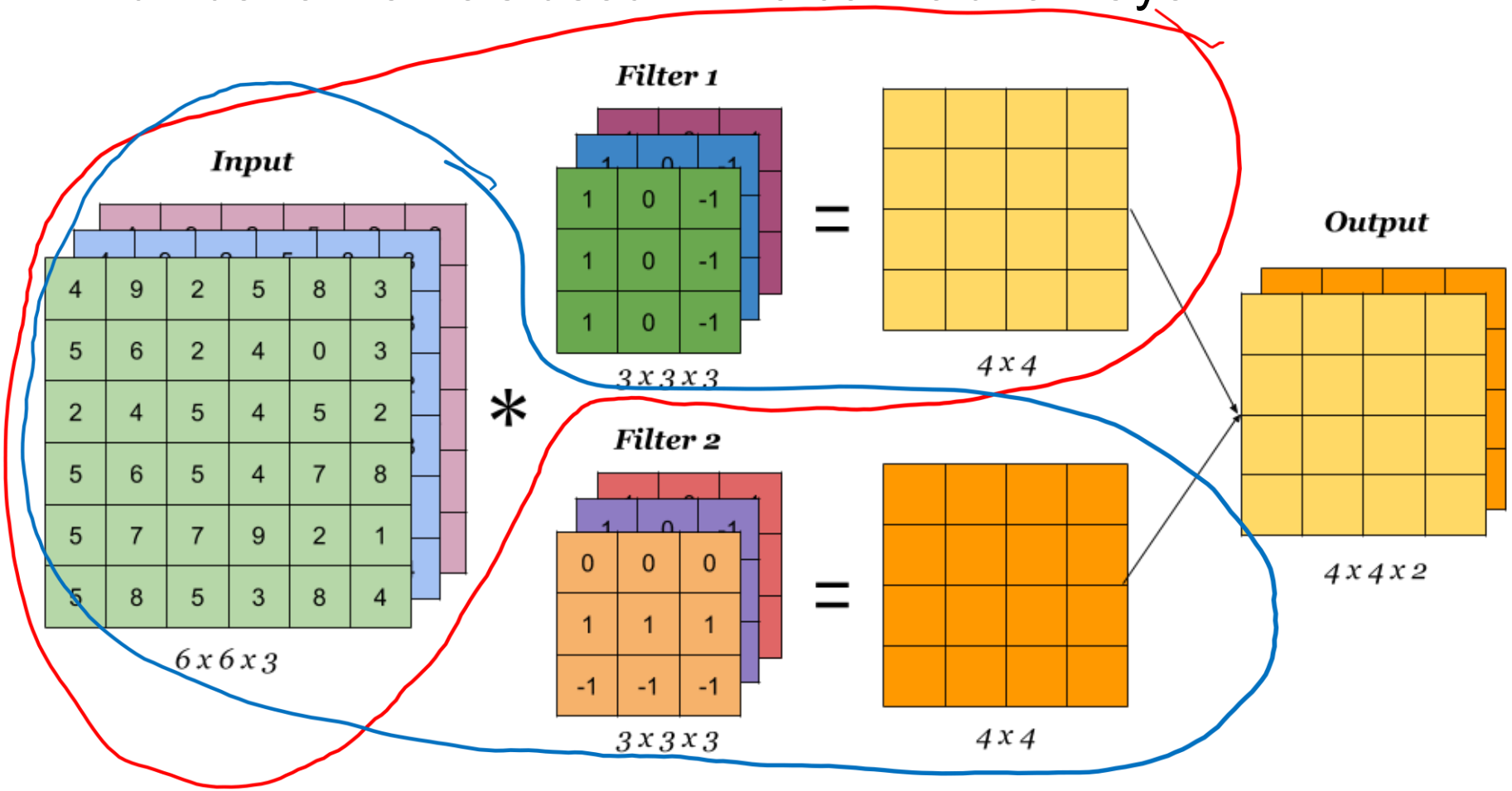
Detecting Multiple Features

Q: What if we want to detect many features of the input? (e.g., both horizontal edges and vertical edges, and maybe even other features?)



Multiple Kernels (Filters)

- The number of outputs (feature maps) is the same as the number of kernels used in the convolution layer.



Example

Input features: 5 × 32 × 32

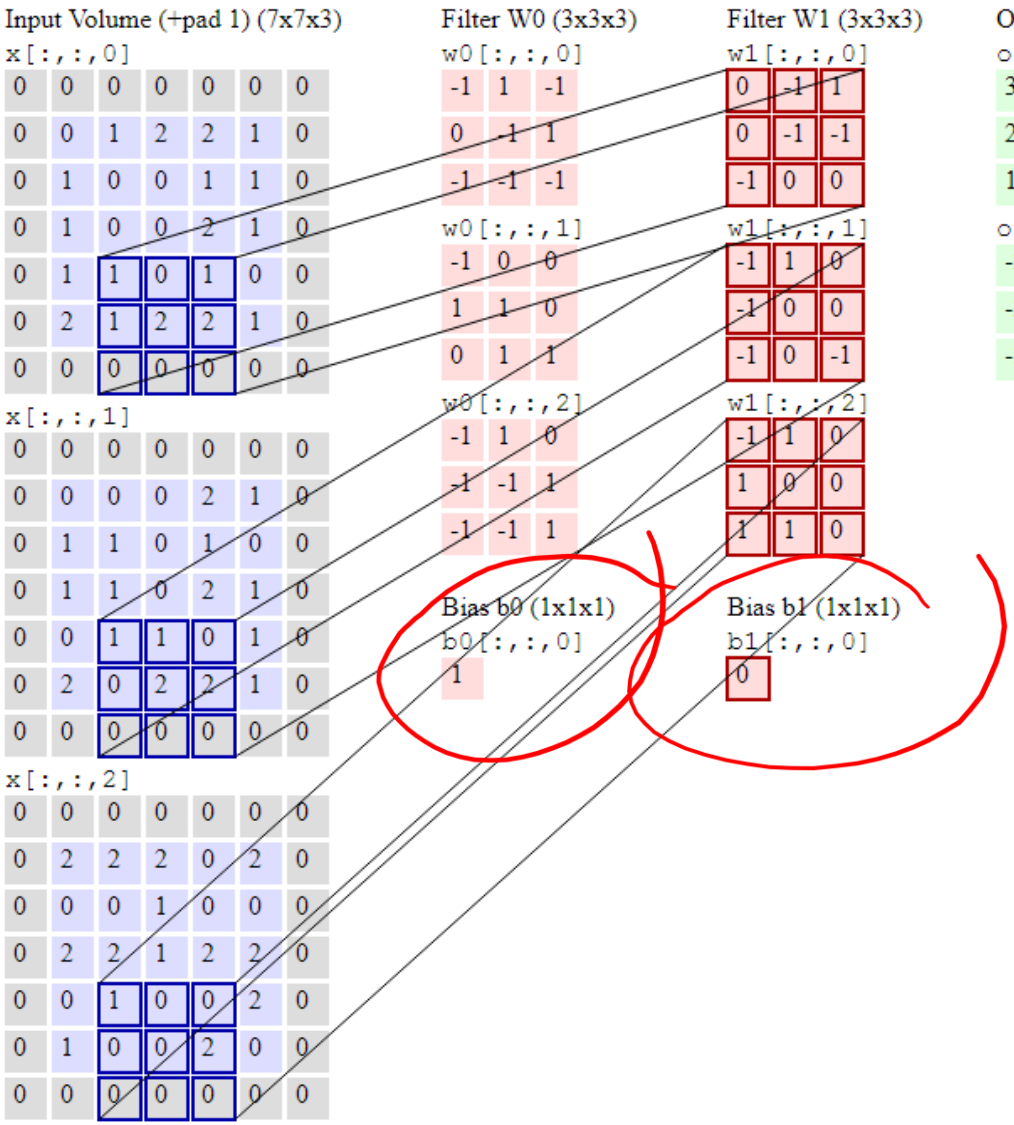
Convolution kernel: 5 × 3 × 3 × 10

#Kernels

Questions:

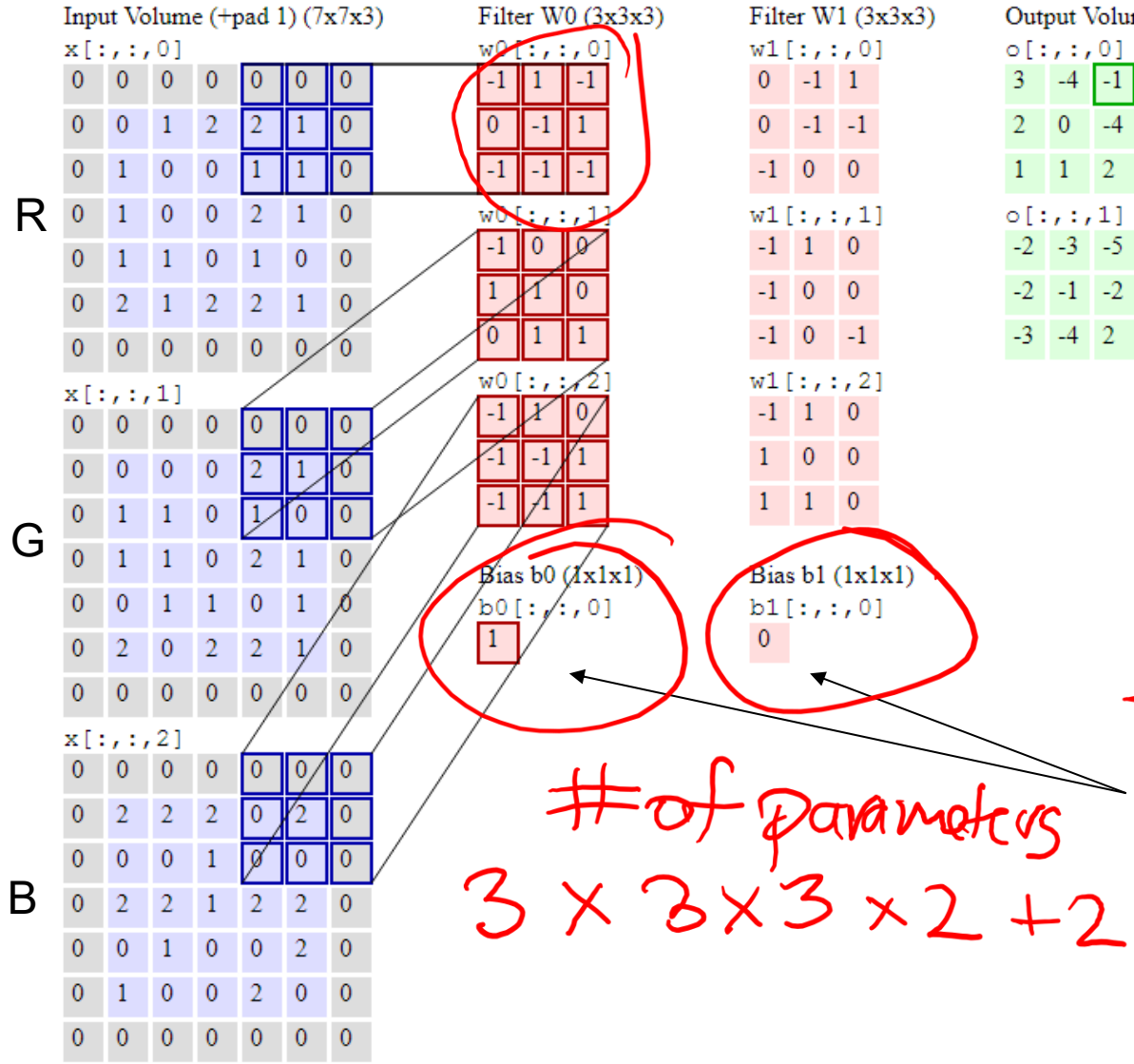
- How many input channels are there? 5
- How many output channels (feature maps) are there? 10
- How many trainable weights are there?

Learning Kernel Parameters



Convolution layer parameters / weights are highlighted in pink and will be learned by the CNN during training

Sample Convolution: Kernel 1



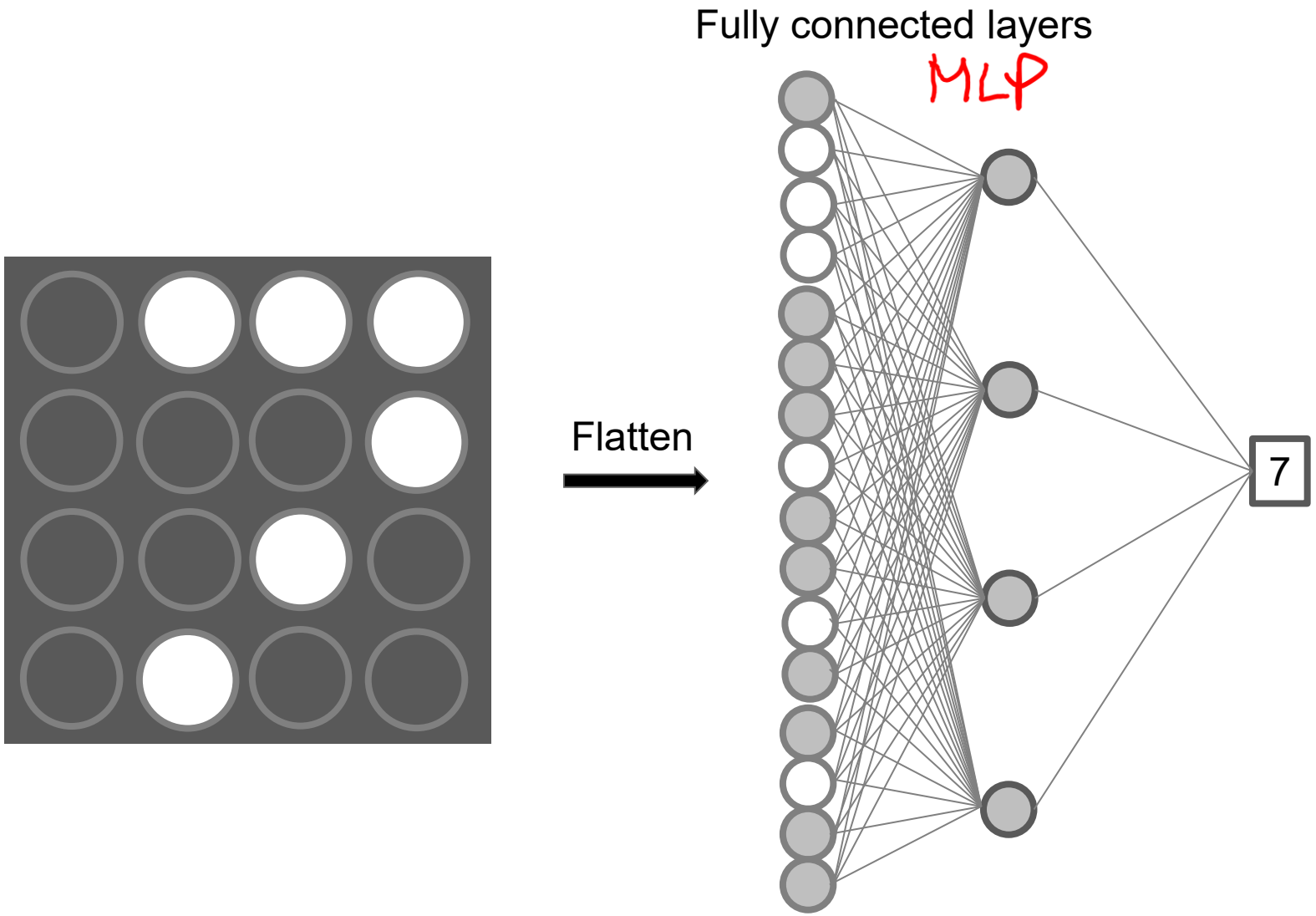
Result of adding 27 numbers + bias

Bias term is often ignored in the example convolutions, but we can assume it's always there and adds a constant to each result

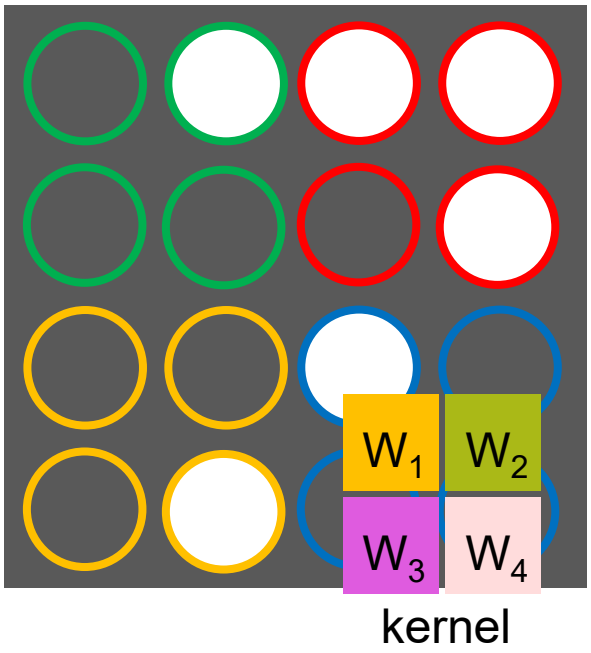
# of parameters

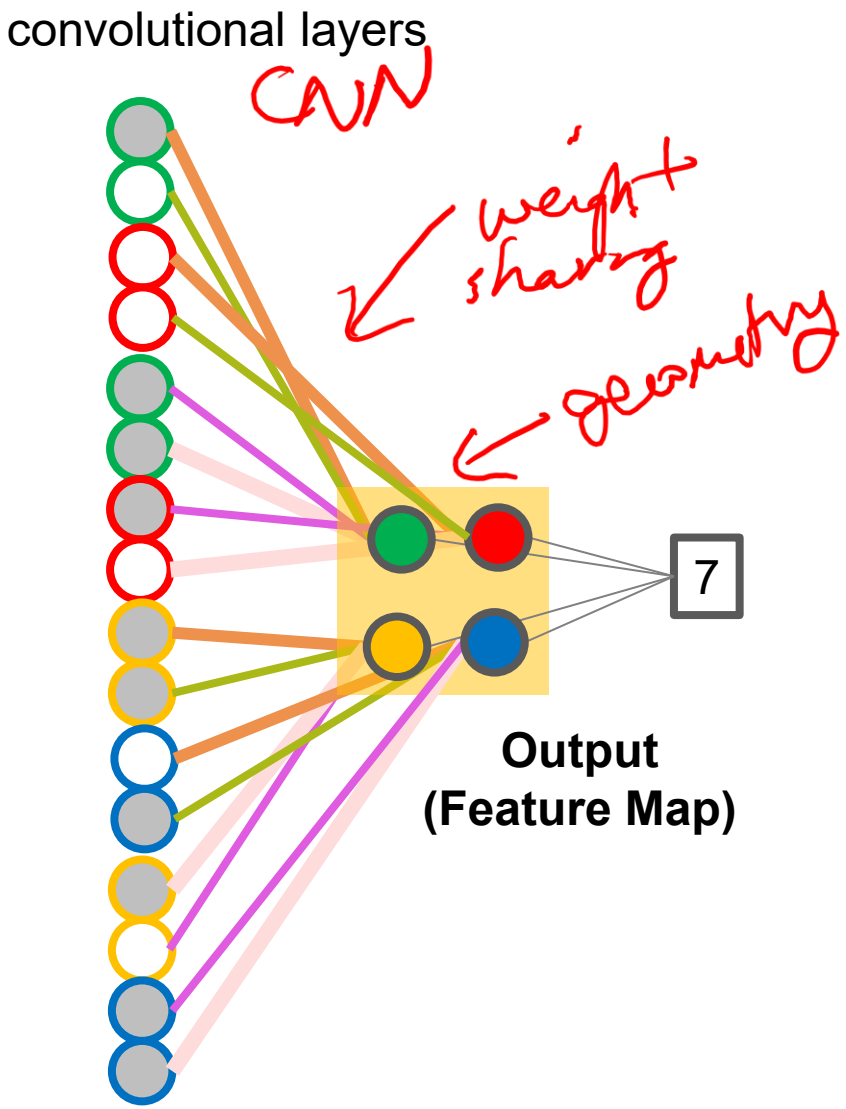
$3 \times 3 \times 3 \times 2 + 2$

Something to think about…



Linear to Convolutional Layers





Summary

- Convolutional operations can be achieved using linear layers!
- Convolutional layers constrain the model to take advantage of spatial features.

Week 4 - Announcements

Assignment 2 Released

- Assignment 2 Preparation
  - Pre_2A – CNN
  - Pre_2B – Transfer Learning
  - Ask questions in Support Session or on Piazza
- Assignment 2 – Hand Gesture Recognition
  - Create your own testing dataset (new data)
  - Dataset has errors – find and correct outliers or leave them and see how it affects performance
  - Large part of the evaluation depends on performance on new data
  - Due Tuesday, Oct 7 at 11:00pm

Projects

- Team Formations
  - Form teams on Quercus
  - Piazza “Search for Teammates!” pinned
  - Project Lists on Shared Google Spreadsheet
  - **1% if done by due date (Tuesday, Sept 30)**
- Project Details
  - Short project description
  - Details and link to dataset(s) you plan on using
  - How you will obtain new samples
  - **1% if done by due date (Tuesday, Oct 7)**

Final Exam Vote

- Results of Final Exam survey

Monday, Dec 1, 8am - 10:30am	12 respondents	29 %
Monday, Dec 1, 1pm - 3:30pm	24 respondents	59 %
Monday, Dec 1, 6pm - 8:30pm	20 respondents	49 %
Wednesday, Dec 3, 8am - 10:30am	15 respondents	37 %
Wednesday, Dec 3, 1pm - 3:30pm	30 respondents	73 %
Wednesday, Dec 3, 7pm - 9:30pm	25 respondents	61 %
Friday, Dec 5, 8am - 10:30am	10 respondents	24 %
Friday, Dec 5, 1pm - 3:30pm	23 respondents	56 %
Friday, Dec 5, 6pm - 8:30pm	24 respondents	59 %

Next Time

Homework

- Review Pre2A Sample Code
- Review Pre2B Sample Code

Q/A Support Session

- Assignment 2
- Project Discussions
- Questions about Sample Code

Lecture Week 5

- CNN Architectures
- Transfer Learning
- Applications